

ECSE 4560/ 6560

Modern Communication Systems

Lecture 10: OFDM Decoding

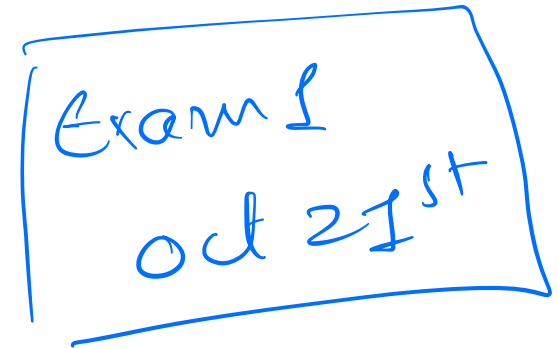


Ish Jain, Fall 2025



Announcements

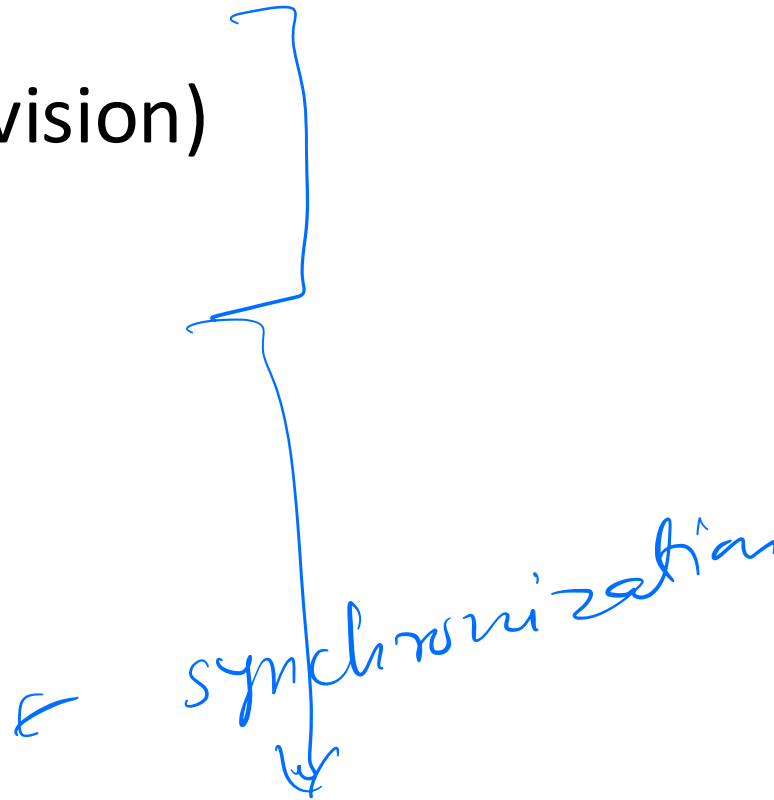
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Decoding**
- Projects
 - Project Topics and Template posted on Piazza
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due today Friday **10/3**
 - 1 page: Project Title, Abstract, and Evaluation Plan
 - **Extra office hour?**
- HW3/Lab3 posted. Due Friday, 10/10



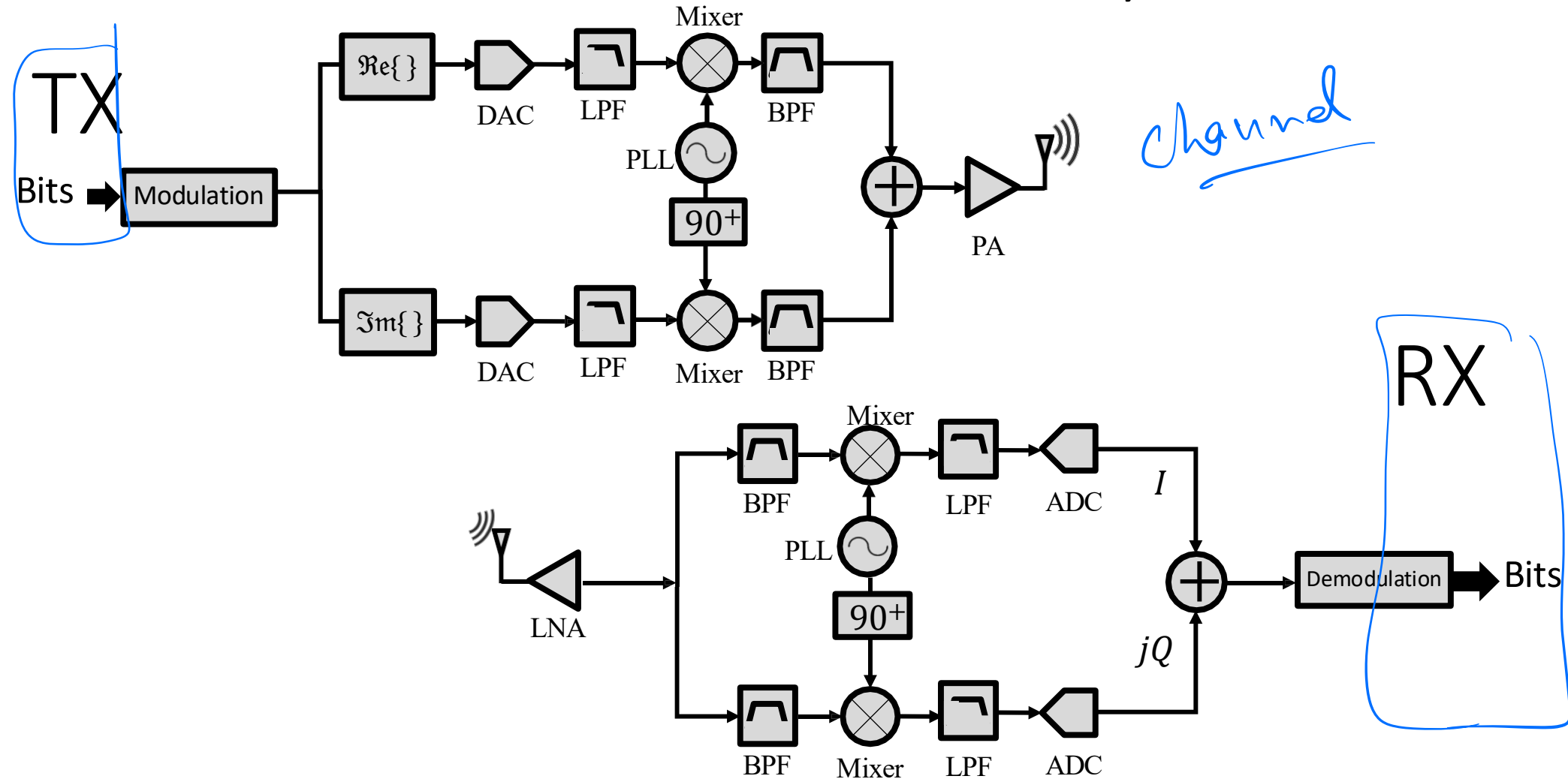
Today's Class: OFDM Decoding

- OFDM

- Orthogonality (revision)
- DFT Encoding
- **Cyclic Prefix**
- **IDFT Decoding**
- **Packet detection**
- **CFO correction**
- Channel Estimation
- SFO and Residual Phase correction



Wireless Communication System



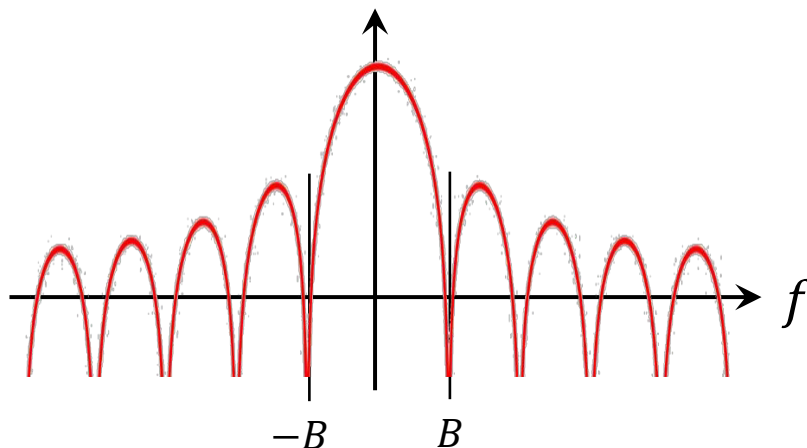
Single Carrier Modulation

Symbols modulated on a single carrier frequency: $s[n]e^{-j2\pi f_c t}$

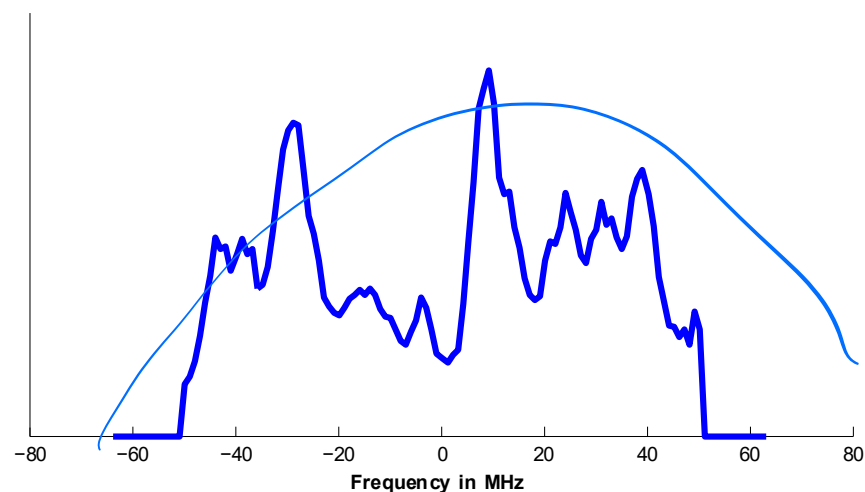
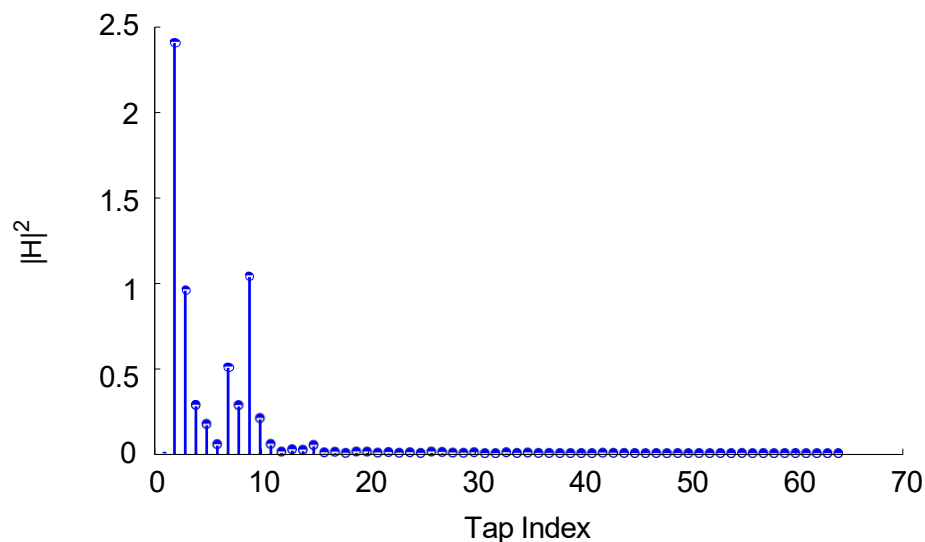
Single Carrier Modulation

Symbols modulated on a single carrier frequency

- Low Spectral Efficiency: sinc & raised cosine leakage



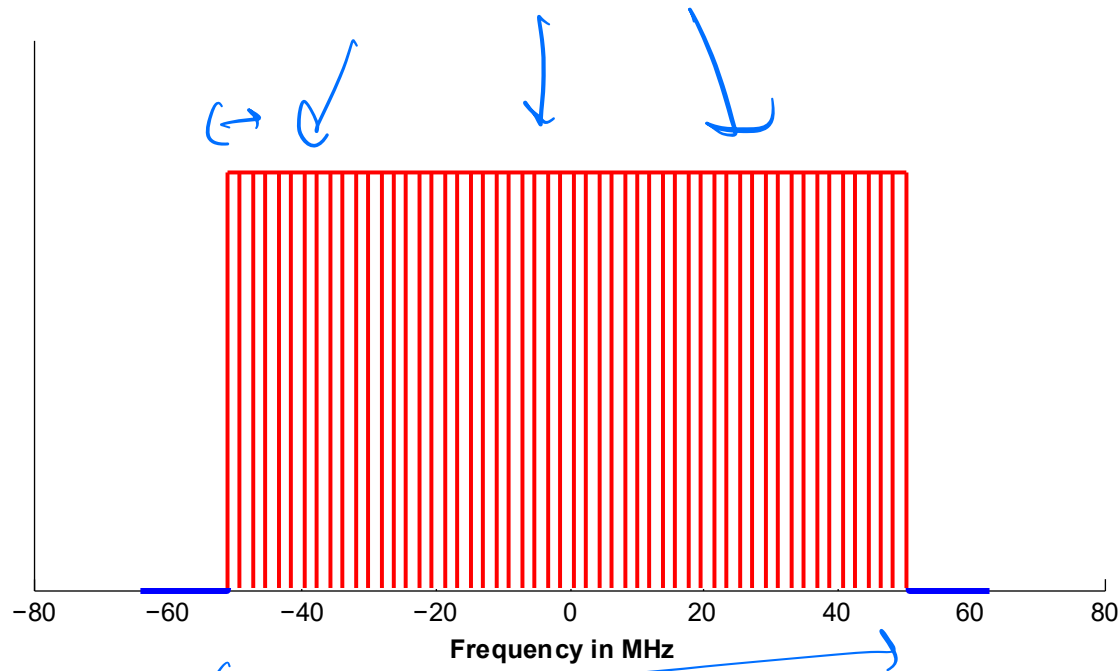
- ISI: Inter-Symbol-Interference limits performance



Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



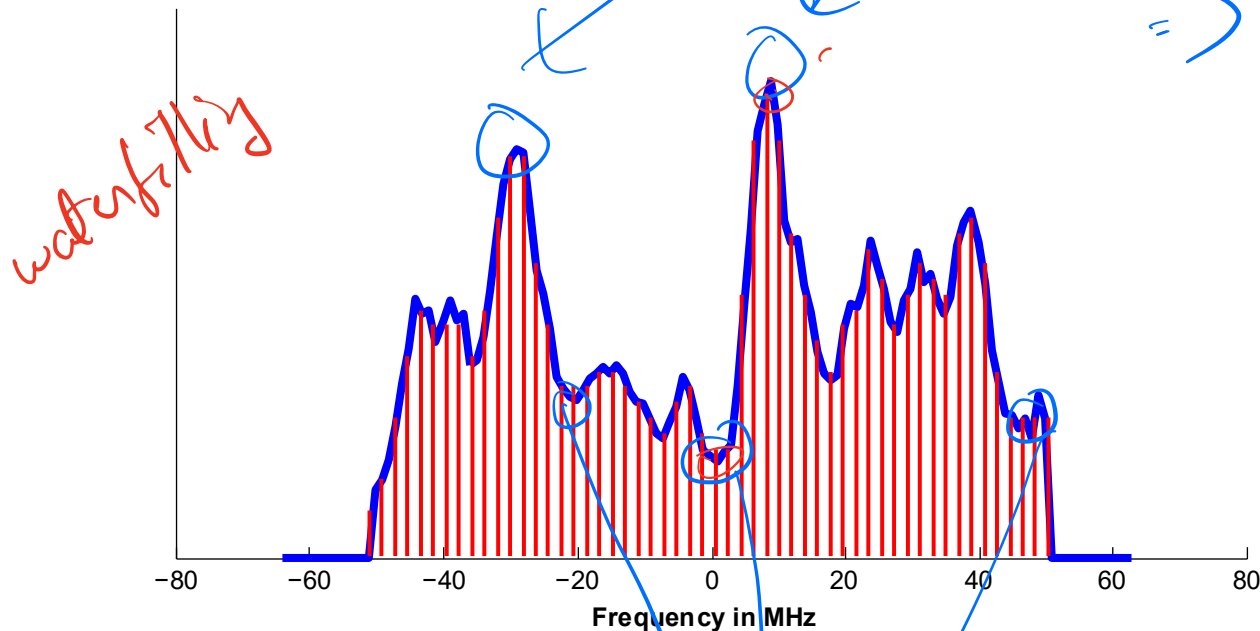
$$x(t) = \sum_i s_i[n] e^{-j2\pi f_i t}$$

- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j2\pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j2\pi f_i t}$$

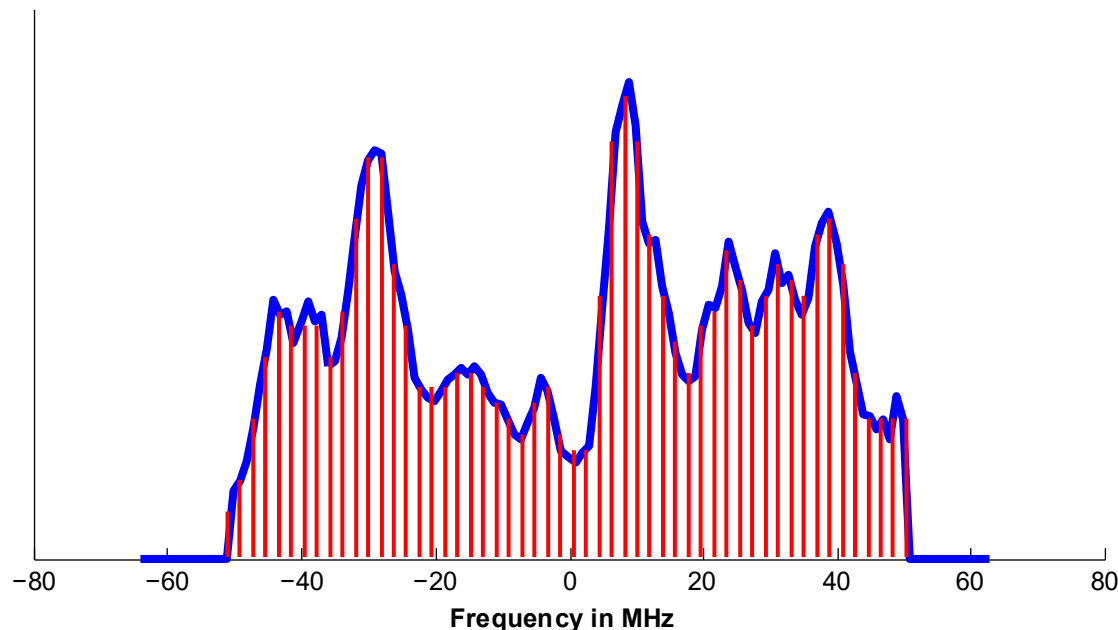
- Transmit symbols on different carriers in narrow bands
- Channel is Flat $\rightarrow$ No need to worry about ISI

bad $\Rightarrow$ *not use these subcarriers*
SNR is too low

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j 2 \pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j 2 \pi f_i t}$$

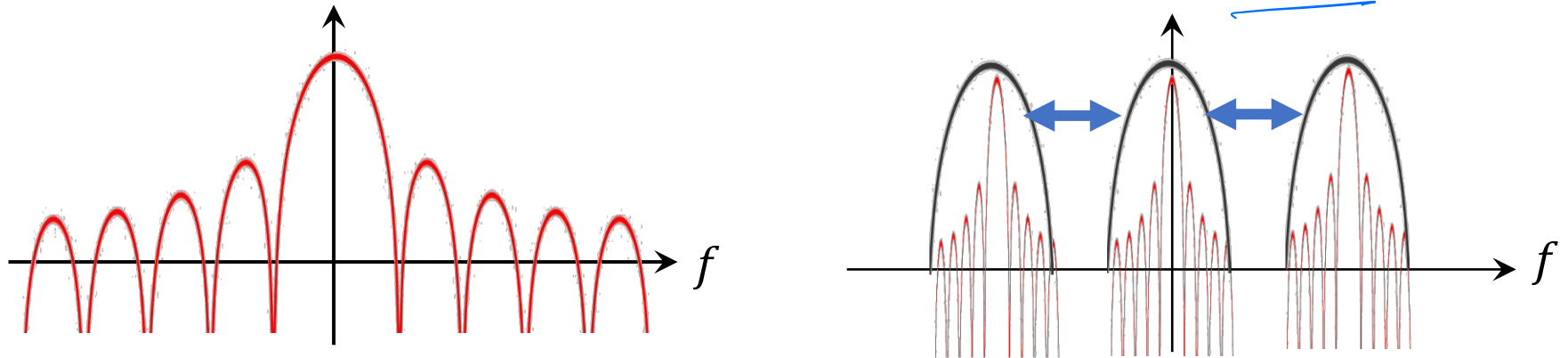
- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Not That Simple!

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



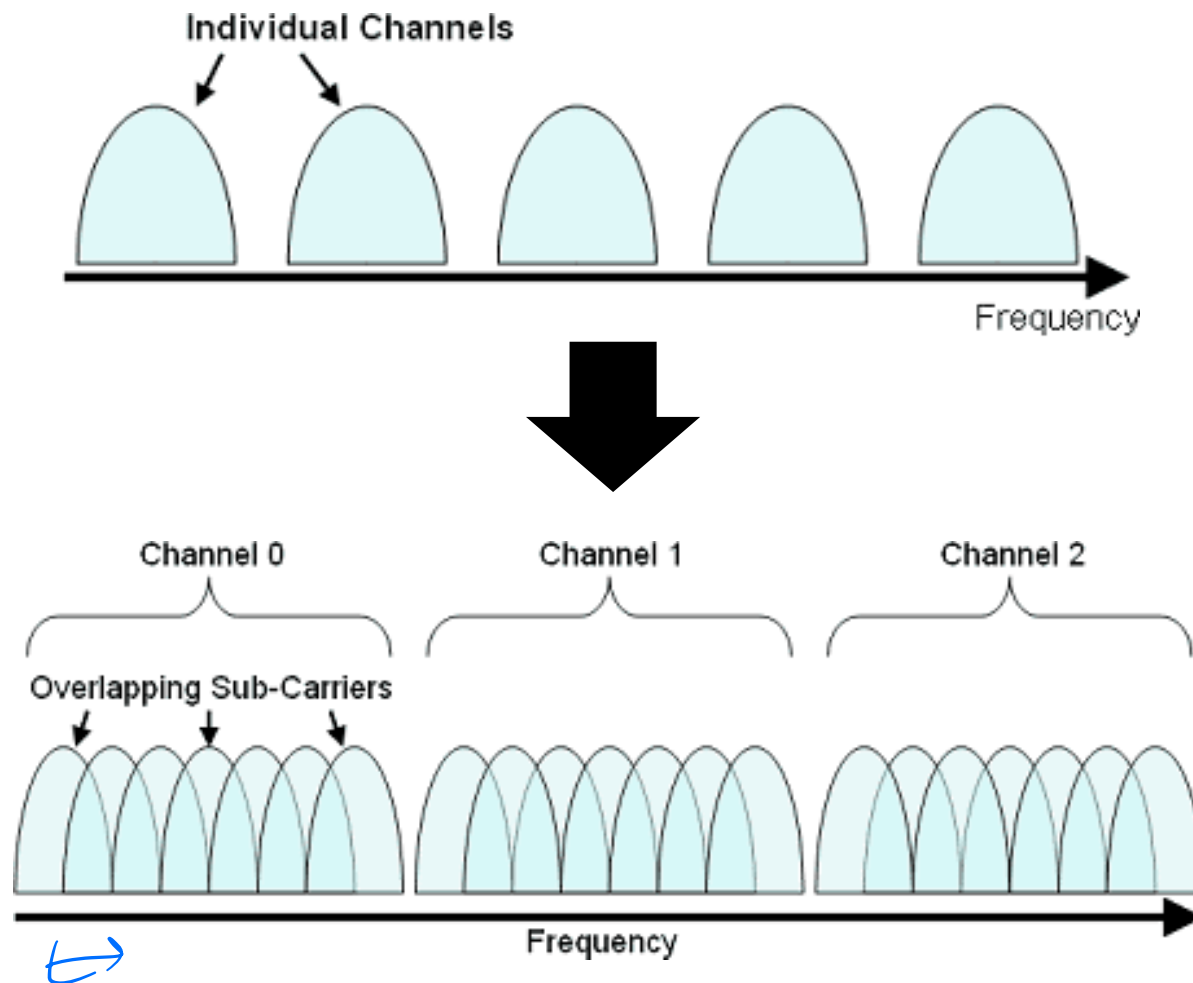
- Significant Leakage between adjacent subcarriers
- Need Guard Bands → Very inefficient!

Solution: Make the Sub-Carriers Orthogonal

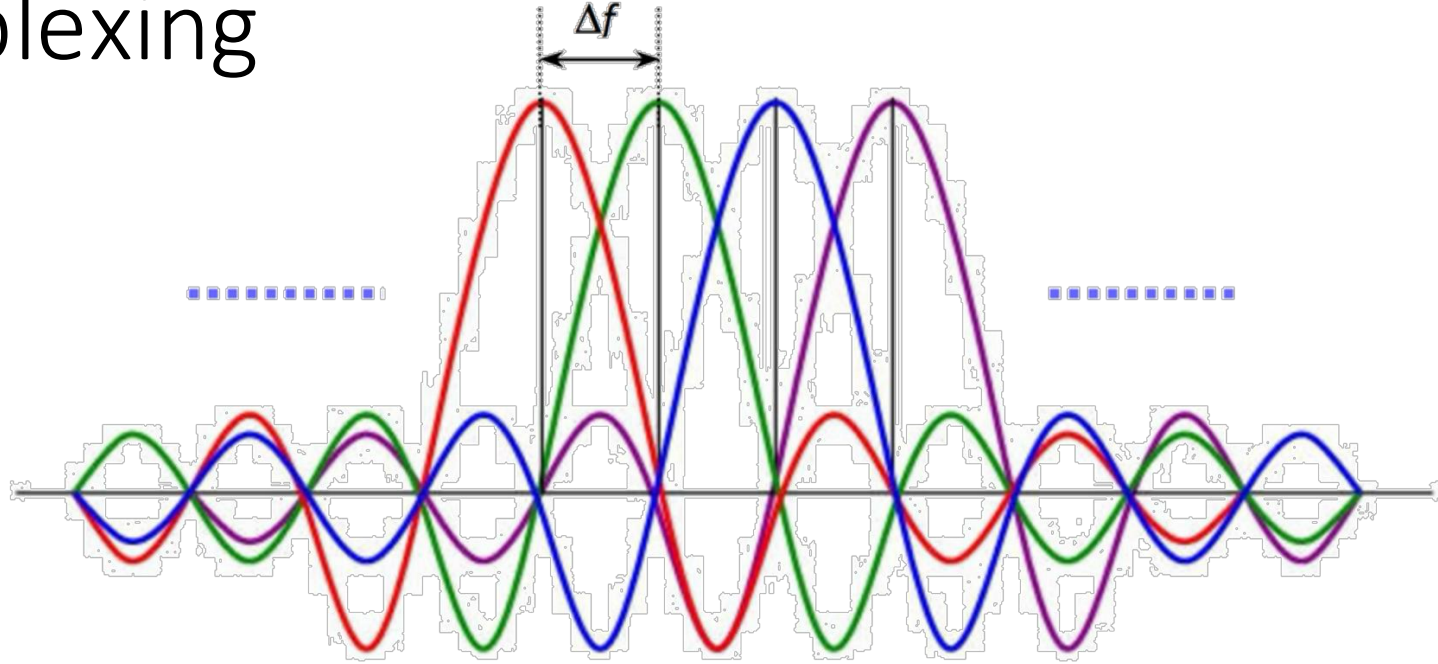
Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

Make the Sub-Carriers Orthogonal



OFDM: Orthogonal Frequency Division Multiplexing

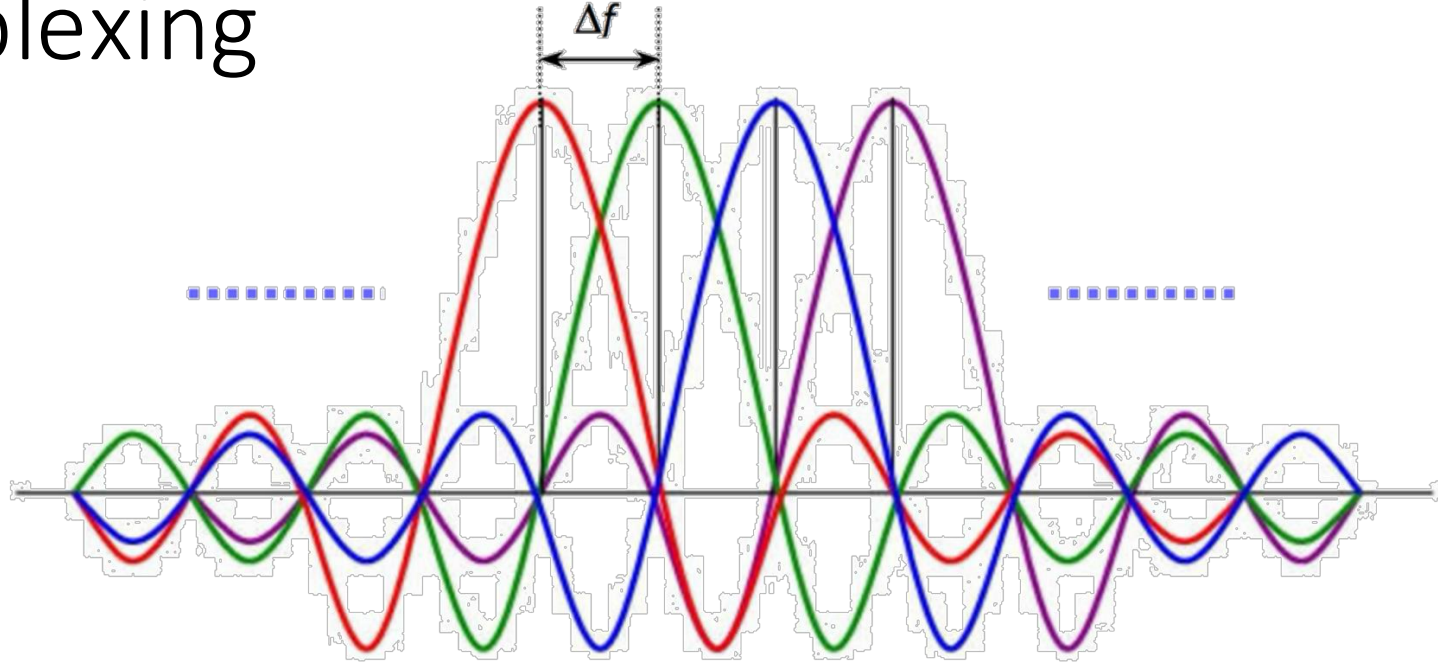


- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
- Subcarrier separation can be very small, for N subcarriers and bandwidth B:

$$\Delta f = \frac{B}{N}$$

total Bw
subcarriers

OFDM: Orthogonal Frequency Division Multiplexing



- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
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How to Achieve This?

OFDM: Orthogonal Frequency Division Multiplexing

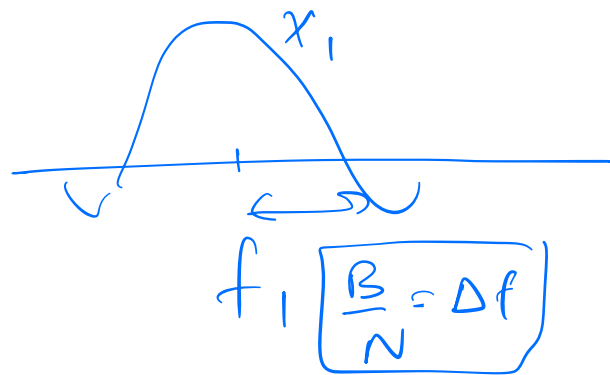
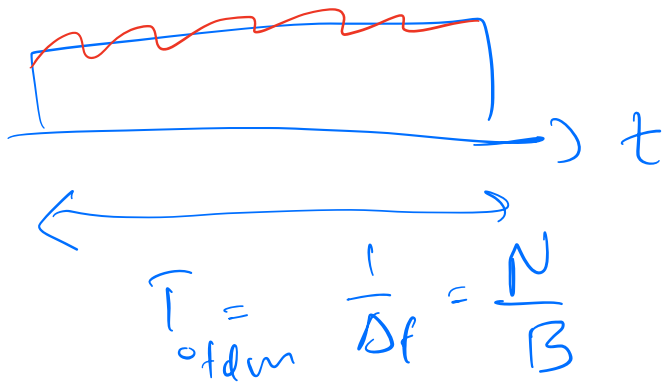
Use DFT: Discrete Fourier Transform

$$\text{N-Point DFT: } X(f_i) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi f_i t}{N}}$$

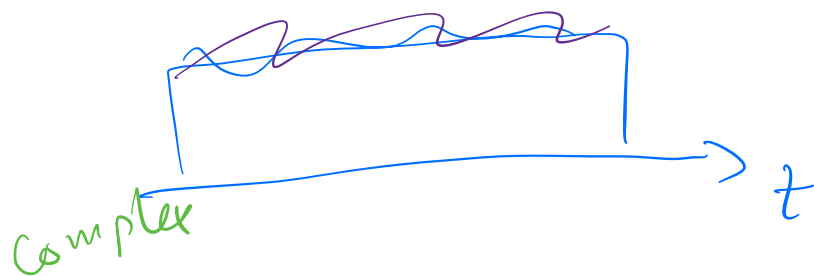
$$\text{N-Point IDFT: } x(t) = \sum_{f_i=0}^{N-1} X(f_i) e^{j \frac{2\pi f_i t}{N}}$$

Send symbols in Frequency Domain

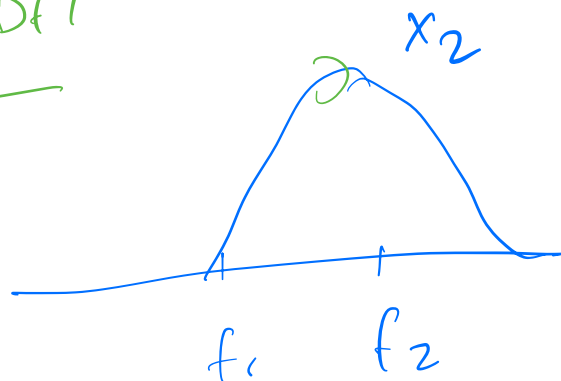
$X(f_i) = s[n] \rightarrow$ Compute and transmit $x(t)$ using IDFT



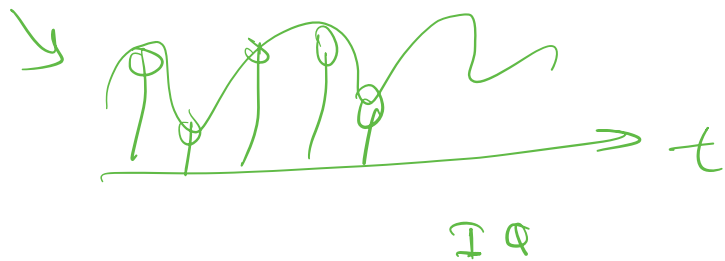
⑦



← IDFT



⑧



OFDM Orthogonality Proof

IDFT Property of DFT

$$y(t) = s(t) = \sum_i X_i e^{j2\pi f_i t}$$

$$\int_0^{T_0} y(t) \left(e^{j2\pi f_l t} \right)^* dt = \frac{B}{N} \int_0^{\frac{N}{B}} \left(\sum_i X_i e^{j2\pi i \frac{B}{N} t} \right) e^{-j2\pi l \frac{B}{N} t} dt$$

DFT
fe
new
freq

$$= \frac{B}{N} \sum_i \int_0^{\frac{N}{B}} X_i e^{j2\pi(i-l) \frac{B}{N} t} dt$$

$i=l$

$$= \underbrace{\frac{B}{N} \int_0^{\frac{N}{B}} X_l dt}_{i=l} + \frac{B}{N} \sum_{i \neq l} \left(\int_0^{\frac{N}{B}} X_i e^{j2\pi(i-l) \frac{B}{N} t} dt \right)$$

$i \neq l$

$$= X_l + \frac{B}{N} \sum_{i \neq l} X_i \underbrace{\int_0^{\frac{N}{B}} e^{j2\pi(i-l) \frac{B}{N} t} dt}_{=0}$$

$$= X_l$$

OFDM: Orthogonal Frequency Division Multiplexing

Send symbols in Frequency Domain

$X(f_Q) = s[n] \rightarrow$ Compute and transmit $x(t)$ using IDFT

- Nsubcarrier $\rightarrow$ IDFT of length N
- Symbols $s[n]$ can come from any modulation: BPSK, QPSK, QAM...
- $x(t)$ is complex $\rightarrow$ need I & Q $\rightarrow$ No point using PAM or ASK ...
- OFDM Symbol: N samples of $x(t)$ generated from the same modulated symbols using IDFT.

- OFDM Symbol Time: $T = N/B$ where B is the bandwidth.

- OFDM Frequency Bin Width: $\Delta f = 1/T = B/N$



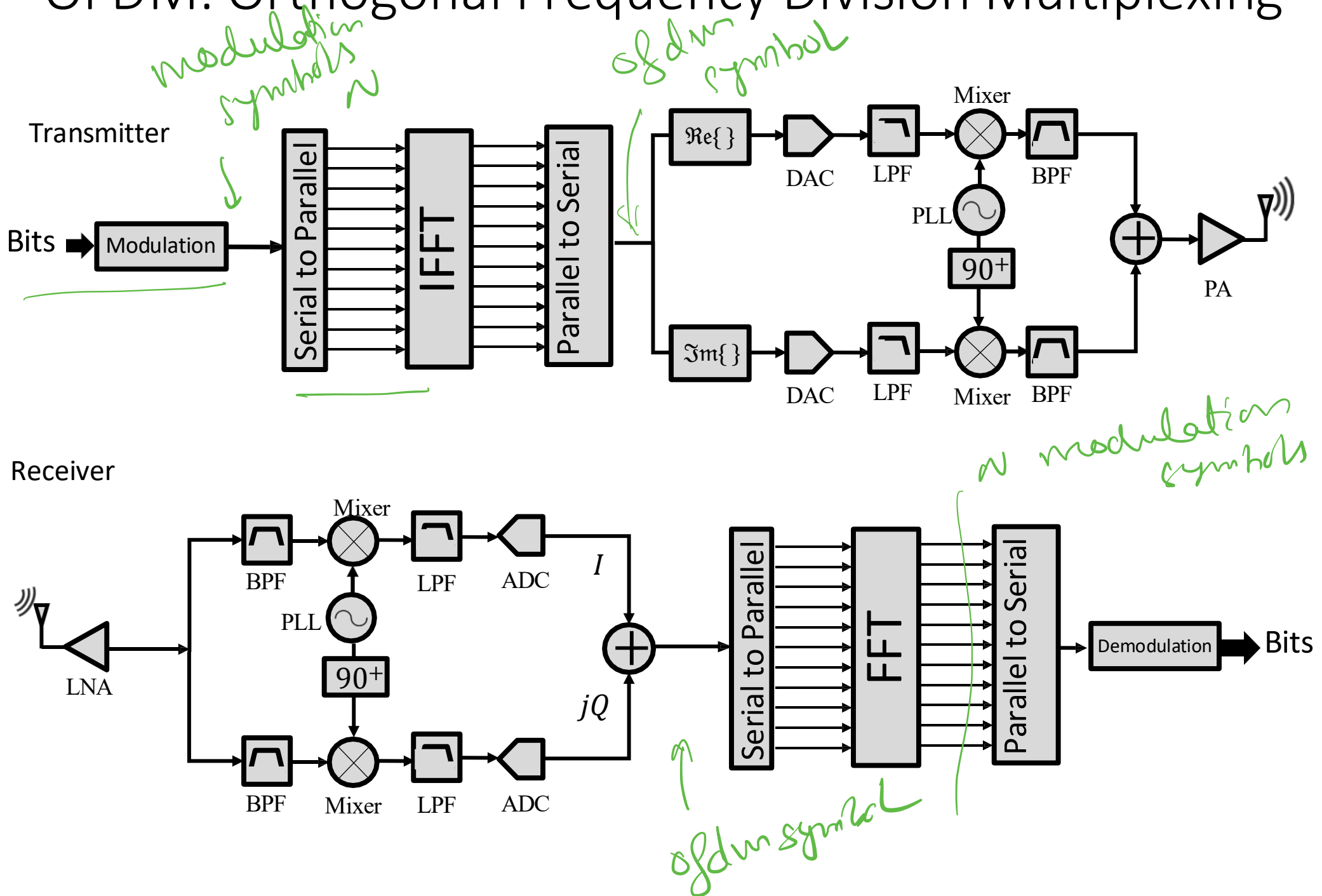
Handwritten notes in green ink:

- ← BPSK symbol
- ↑ N BPSK symbols across freq.

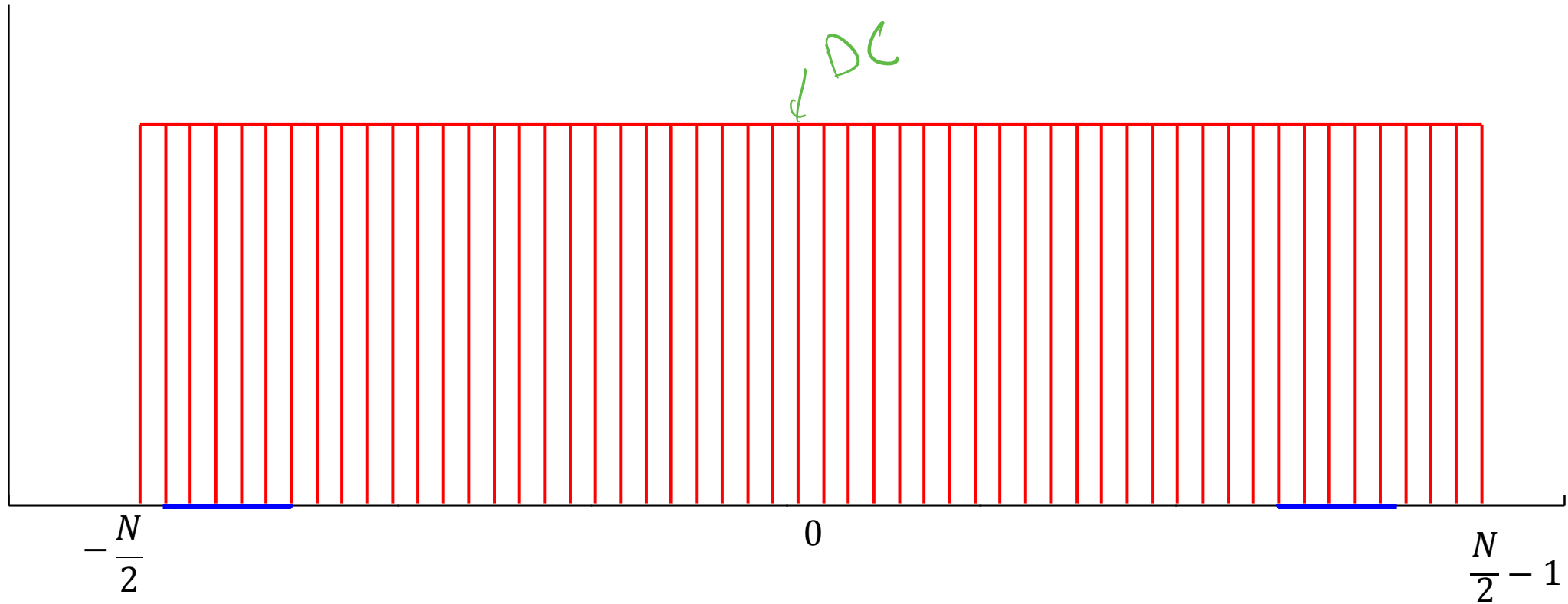
$$T_{\text{ofdm}} = \frac{N}{B}$$

1 ofdm symbol
 $\Rightarrow$ N samples in time

OFDM: Orthogonal Frequency Division Multiplexing



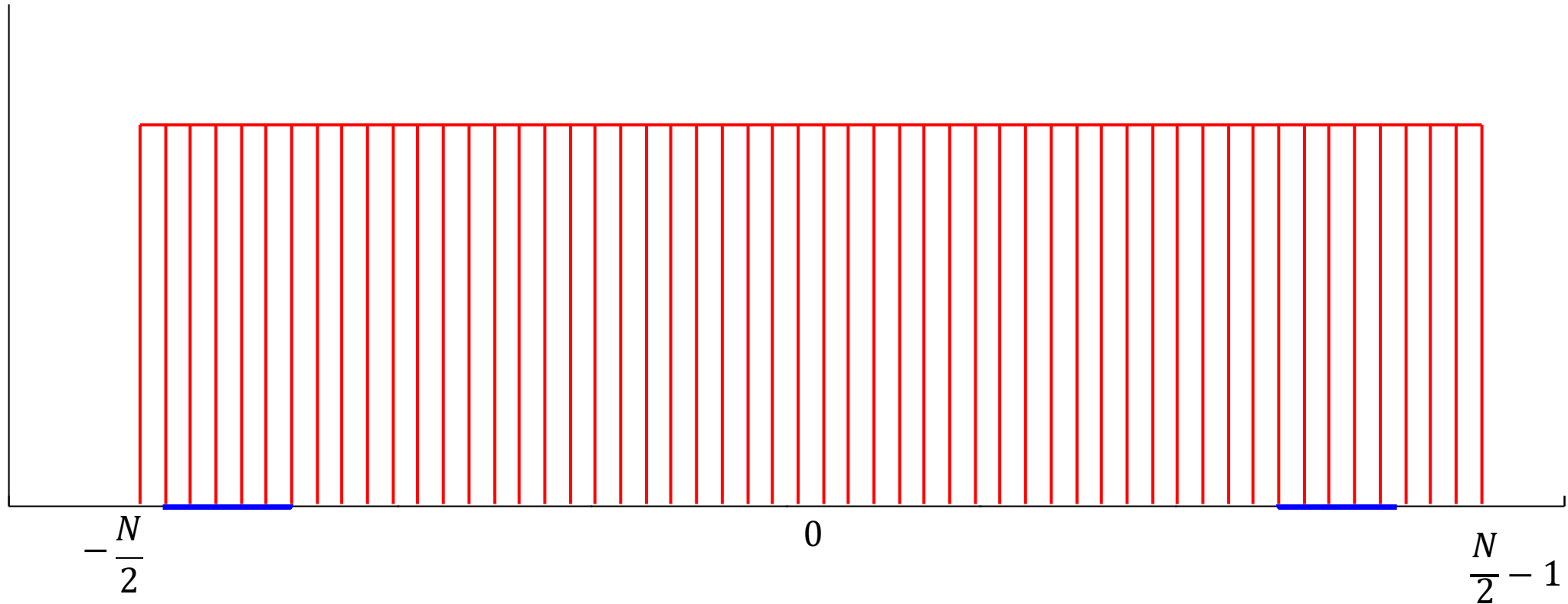
OFDM Symbol in Frequency Domain



- FFT can be represented 0 to $N - 1$ or $-N/2$ to $N/2 - 1$.
- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC

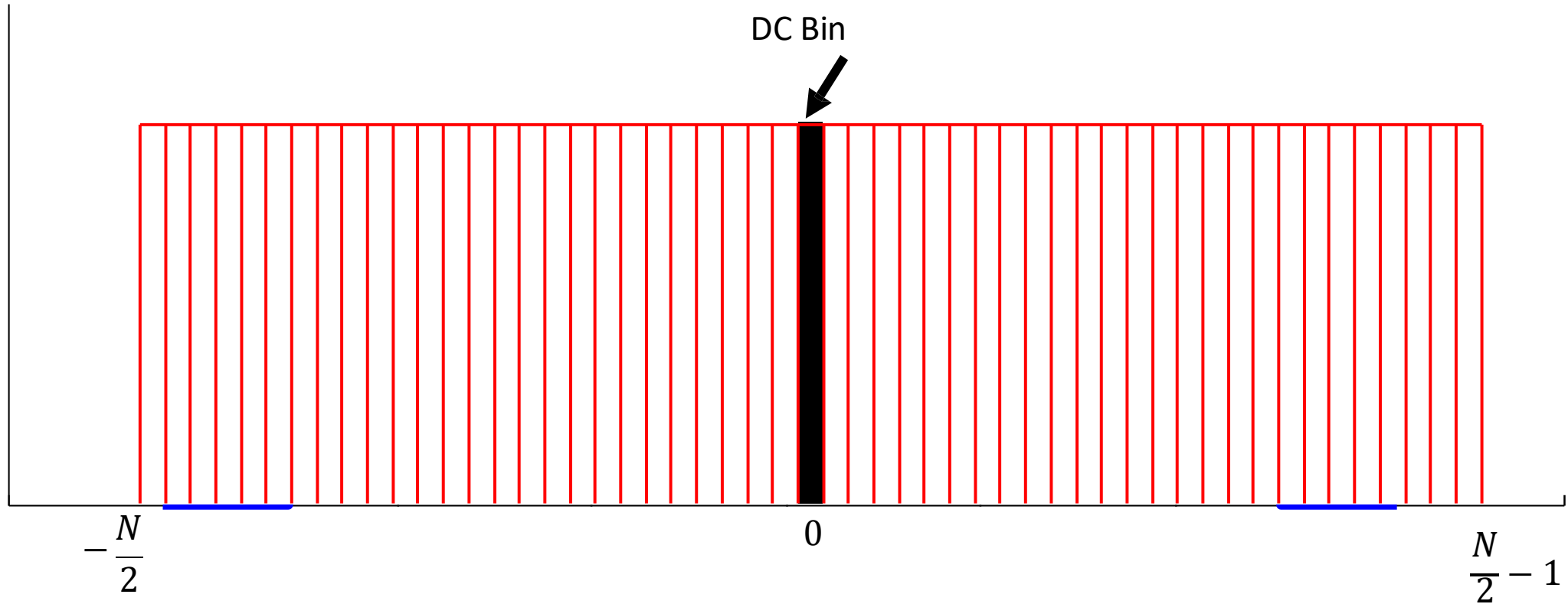
$$X(0) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi 0 t}{N}} = \frac{1}{N} \sum_{t=0}^{N-1} x(t) = DC$$

OFDM Symbol in Frequency Domain



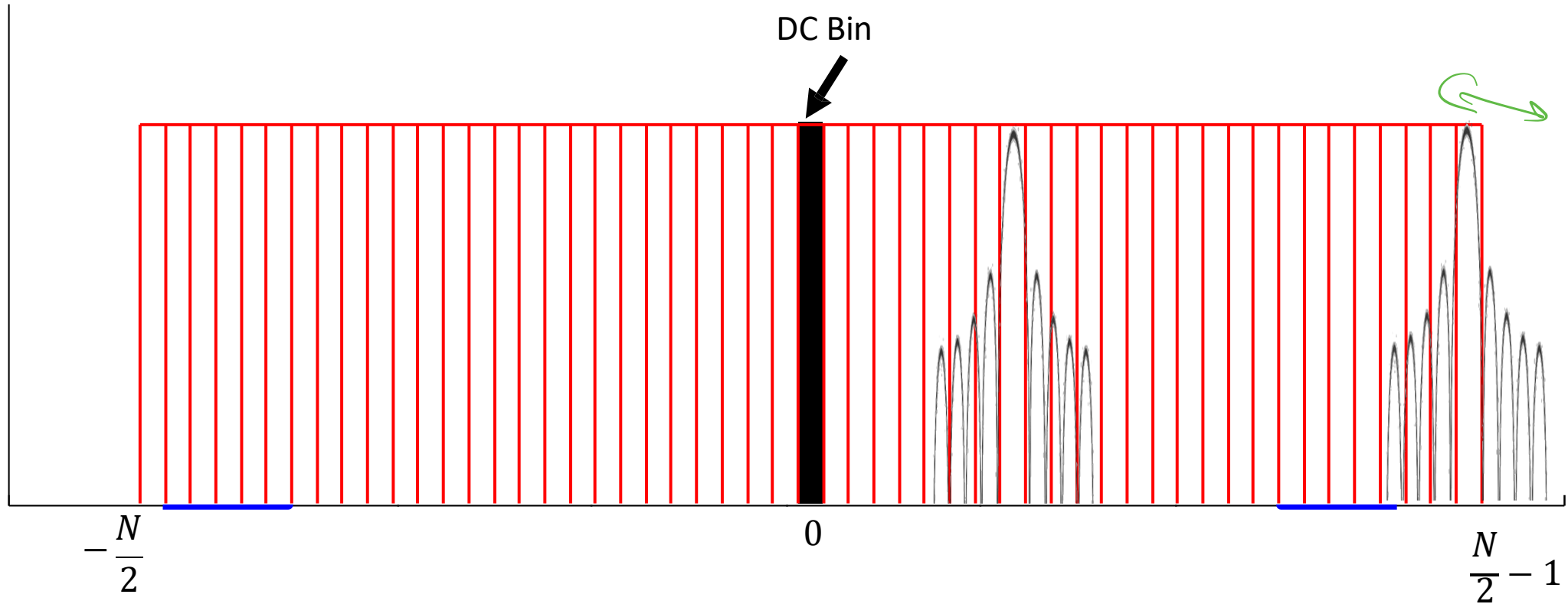
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- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC
- DC of the circuits corrupts bits sent on the 0 bin $\rightarrow$ Do not use 0 bin

OFDM Symbol in Frequency Domain



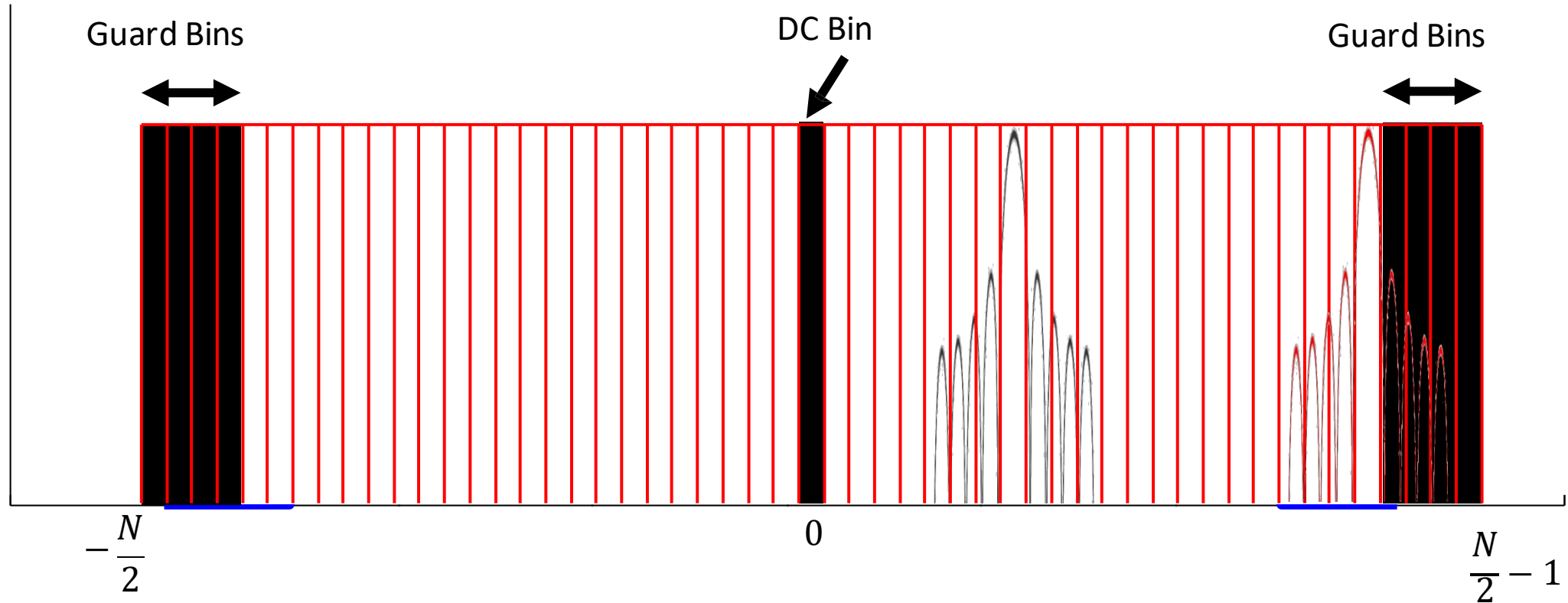
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OFDM Symbol in Frequency Domain



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there

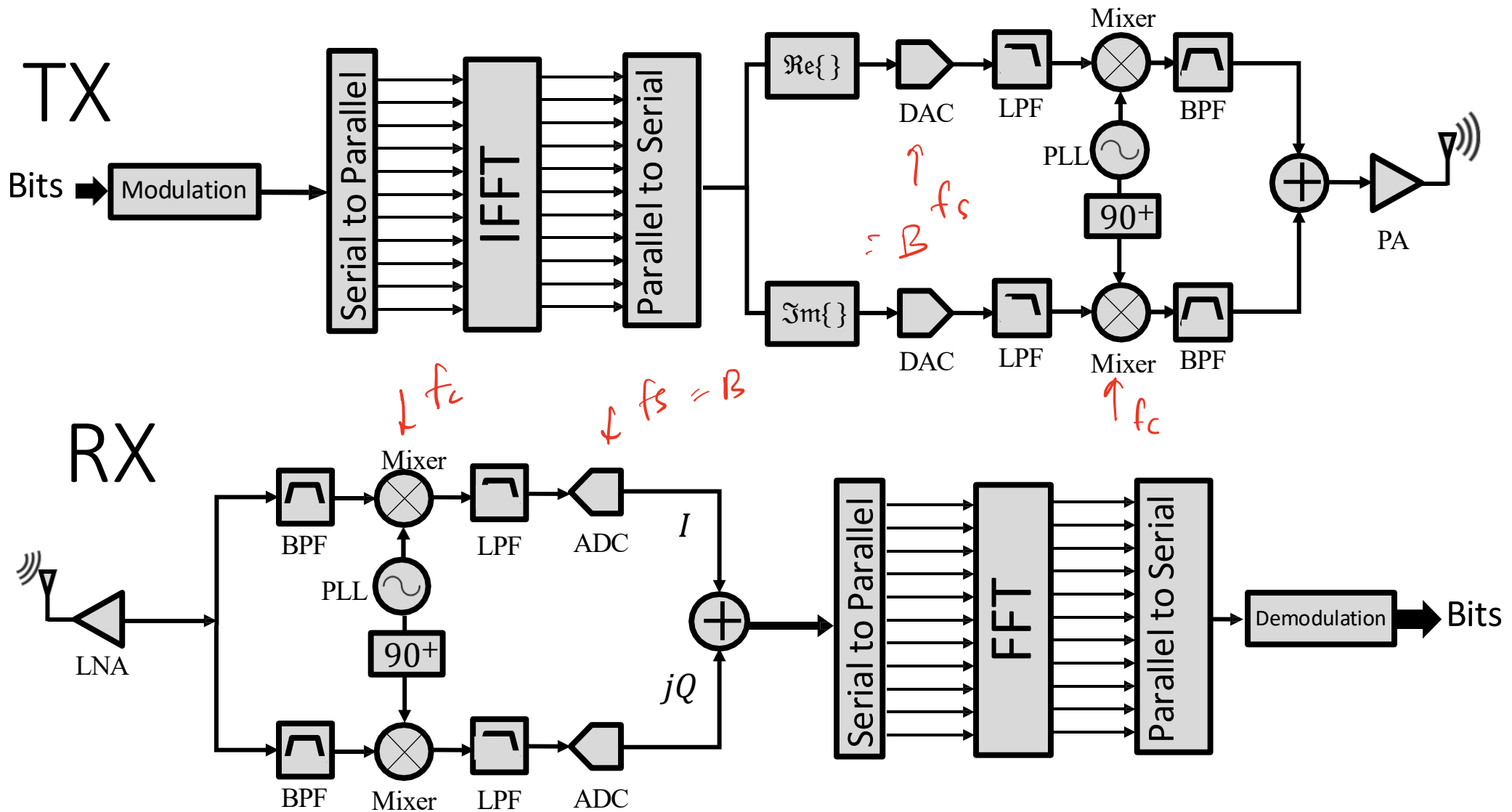
OFDM Symbol in Frequency Domain



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there
- Reduce Number of Guard band from N to 2 → Very Spectrally Efficient

OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers



OFDM Symbol

(coding sche)
↓

Bits:

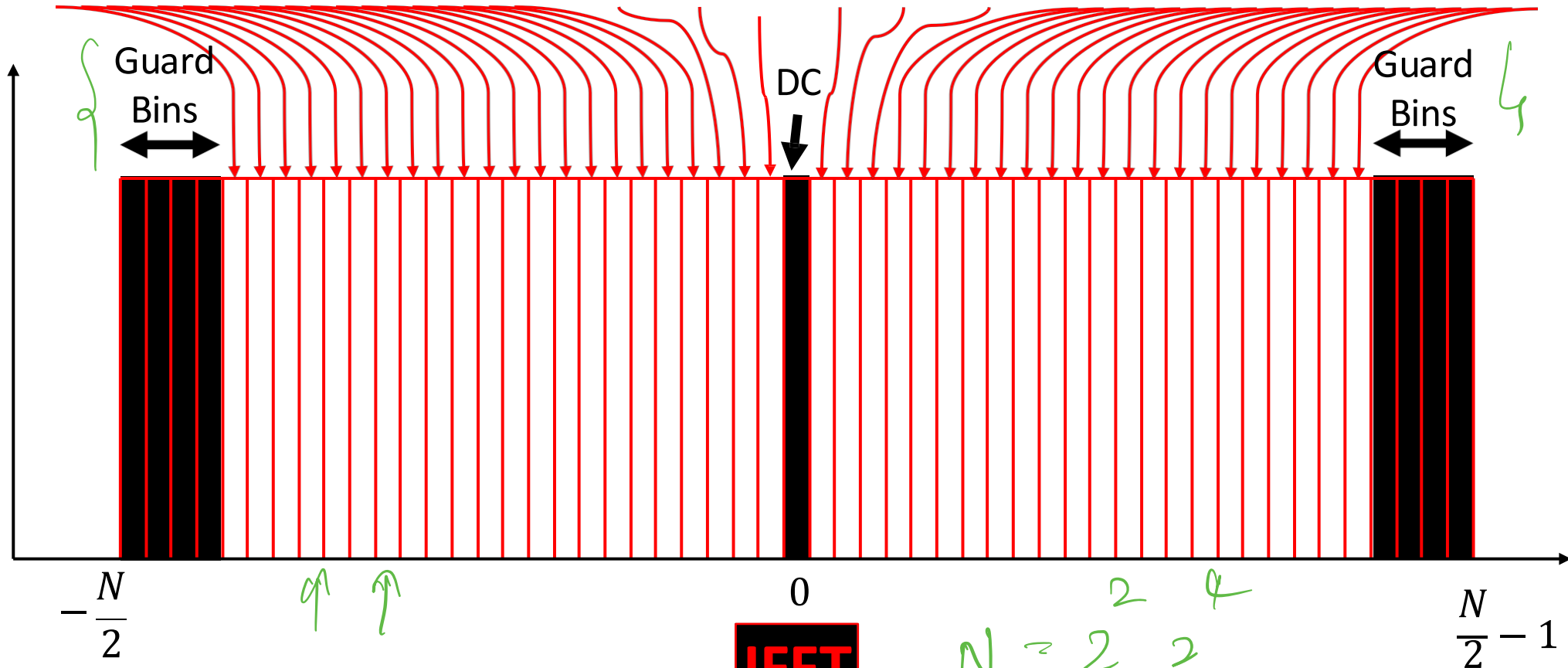
1 0 1 0 1 0 0 0 1 1 0 1 1 0 0

64
128



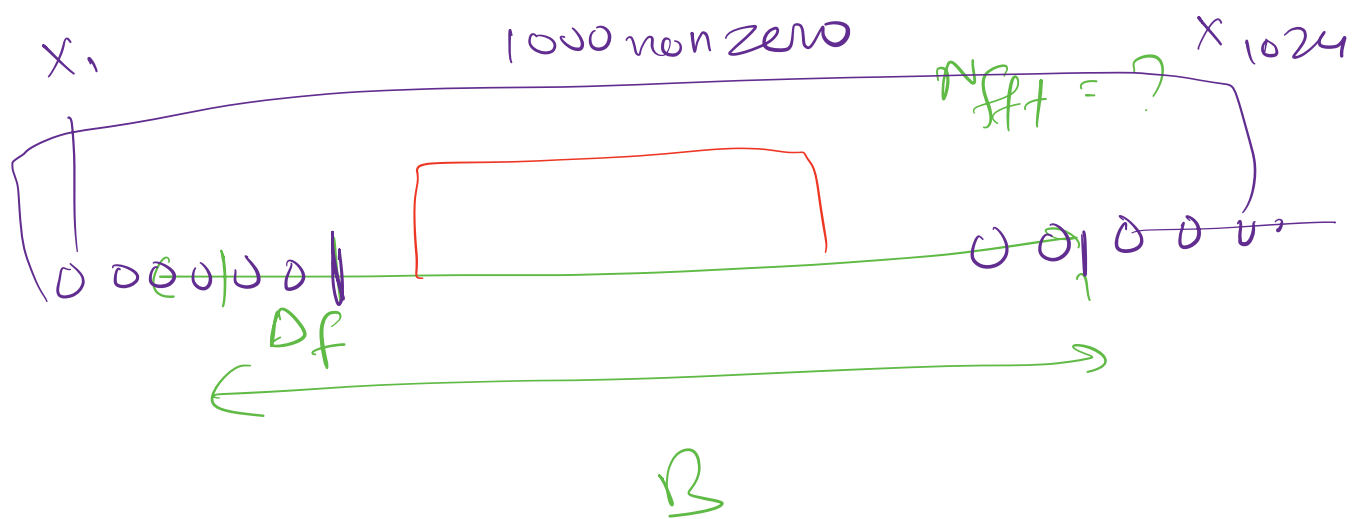
100

..., +1, -1, +1, -1, +1, -1, -1, -1, +1, +1, -1, +1, +1, -1, -1, ...



IFFT

Symbol in Time



$$\Delta f = \frac{B}{N}$$

$$\Delta f = 15 \text{ kHz}$$

$$B = 15 \text{ MHz}$$

$$N = \frac{B}{\Delta f}$$

$$= \frac{15 \text{ M}}{15 \text{ K}} \cdot \frac{1000}{1000}$$

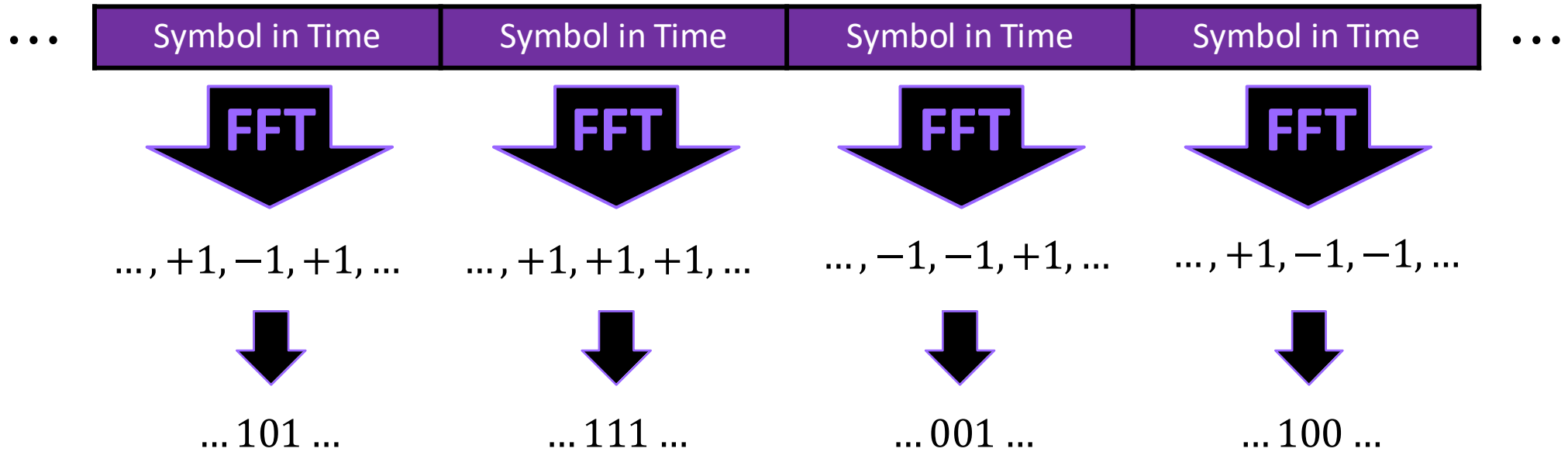
$$= 1024$$

$$N_{\text{fft}} = 2^{10} = 1024$$

with zero padding

OFDM Symbol

→ time



Not That Simple

channel

OFDM Symbol



← FFT Window →



if perfect window is known

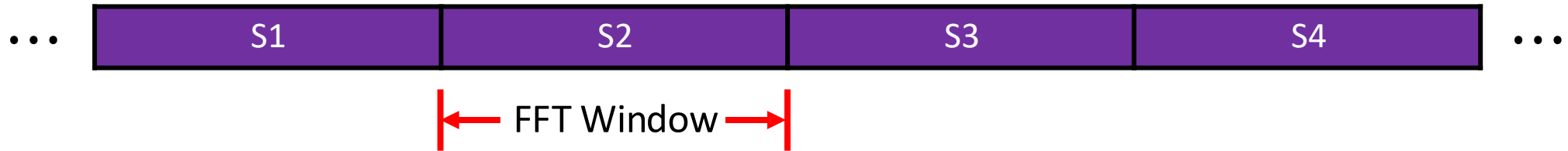
..., +1, -1, +1, ...



... 101 ...

⇒ guess

OFDM Symbol



..., +1, -1, +1, ...



... 101 ...

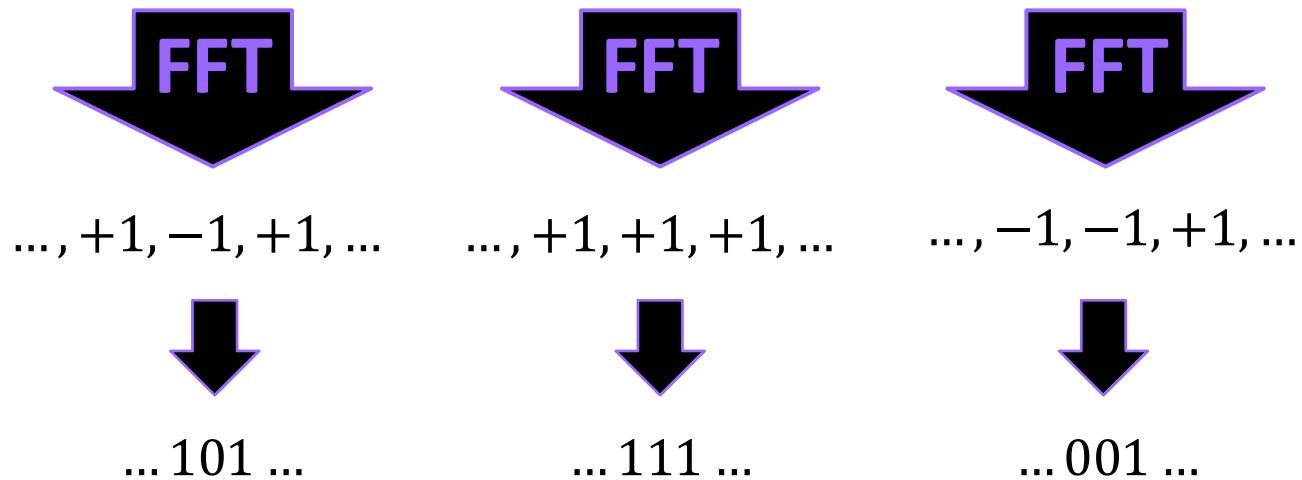
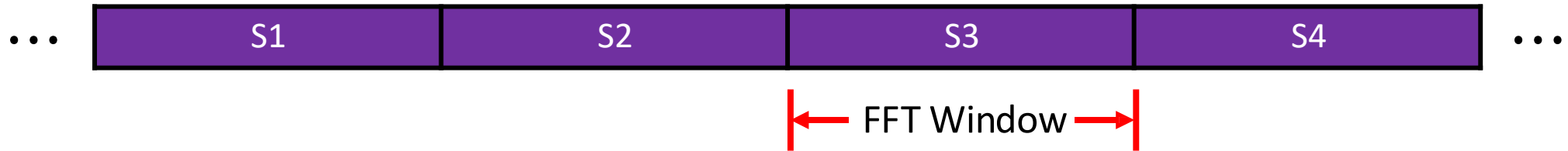


..., +1, +1, +1, ...

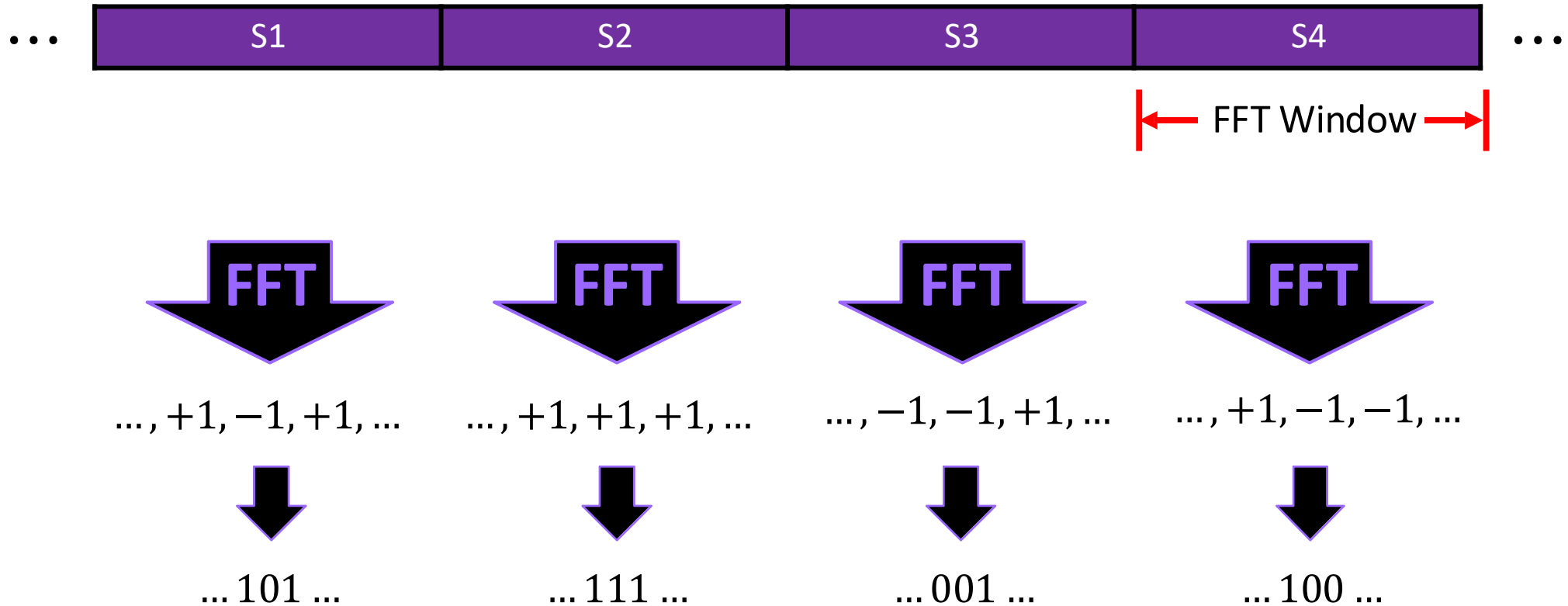


... 111 ...

OFDM Symbol

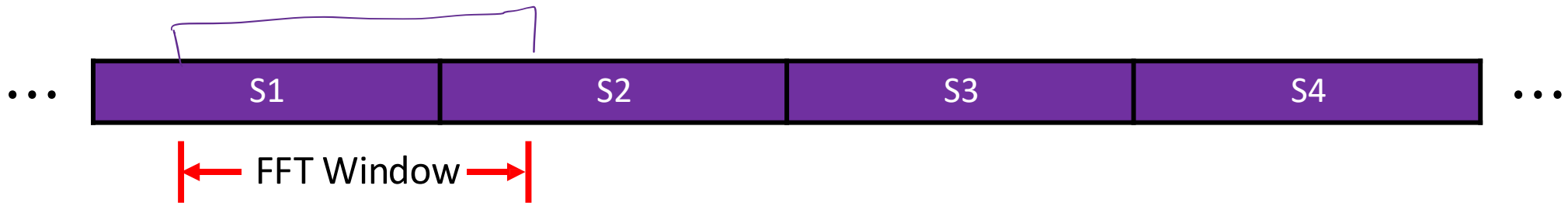


OFDM Symbol



Assumes FFT window is perfectly aligned with symbol boundaries

OFDM Symbol



..., $+0.5 + 1i$, $-0.7 + 0.3i$, ...

✗ Cannot decode!

FFT window is misaligned with symbol

Subcarriers are no longer orthogonal.

OFDM Cyclic Prefix



- DFT (FFT) assumes time samples are periodic of period N

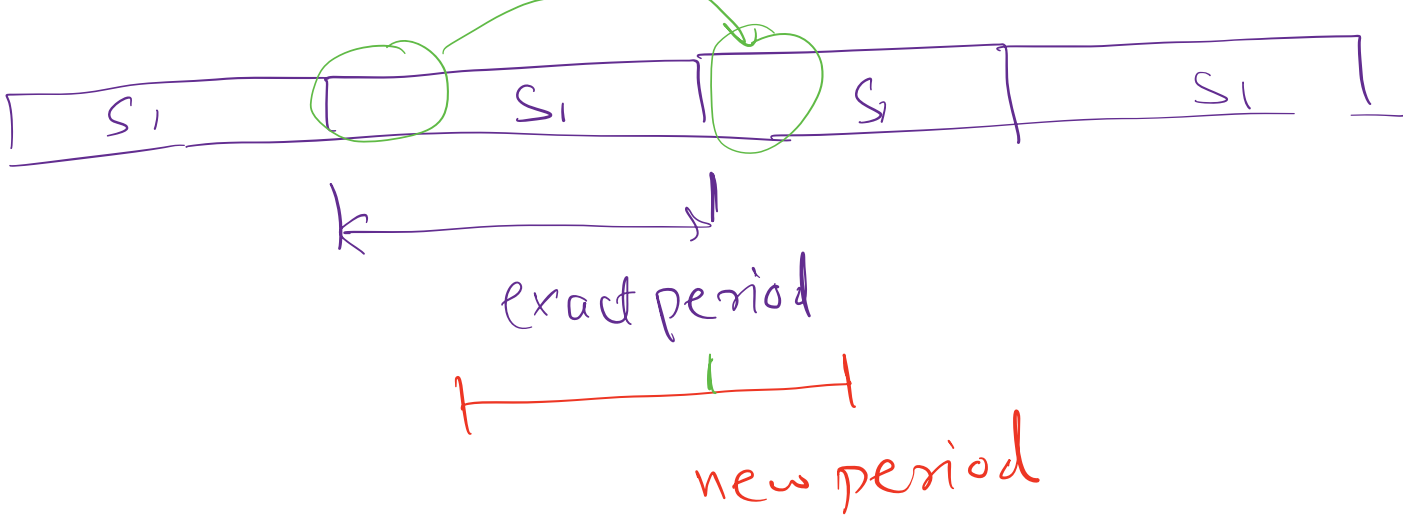
- Circular Shift before taking FFT:

$$x[t] \rightarrow X[f]$$

$$x[t - \tau \bmod N] \rightarrow X[f] e^{-j \frac{2\pi f \tau}{N}}$$

cyclic shift
by τ

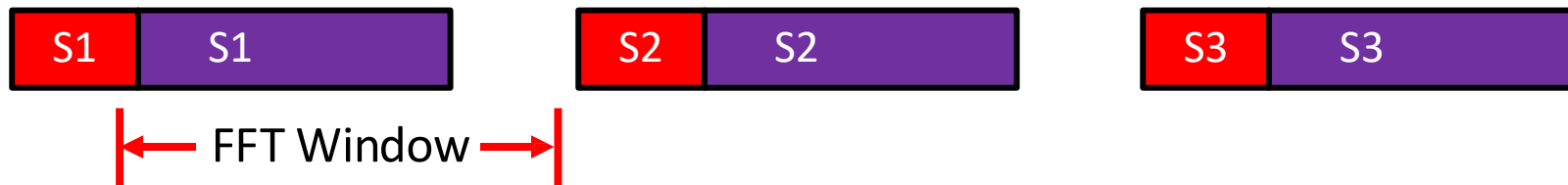
+ addition
phase term
that can be compensated
in channel equalizer



↓ Decoding DFT

cyclic shifted version of S_1

OFDM Cyclic Prefix

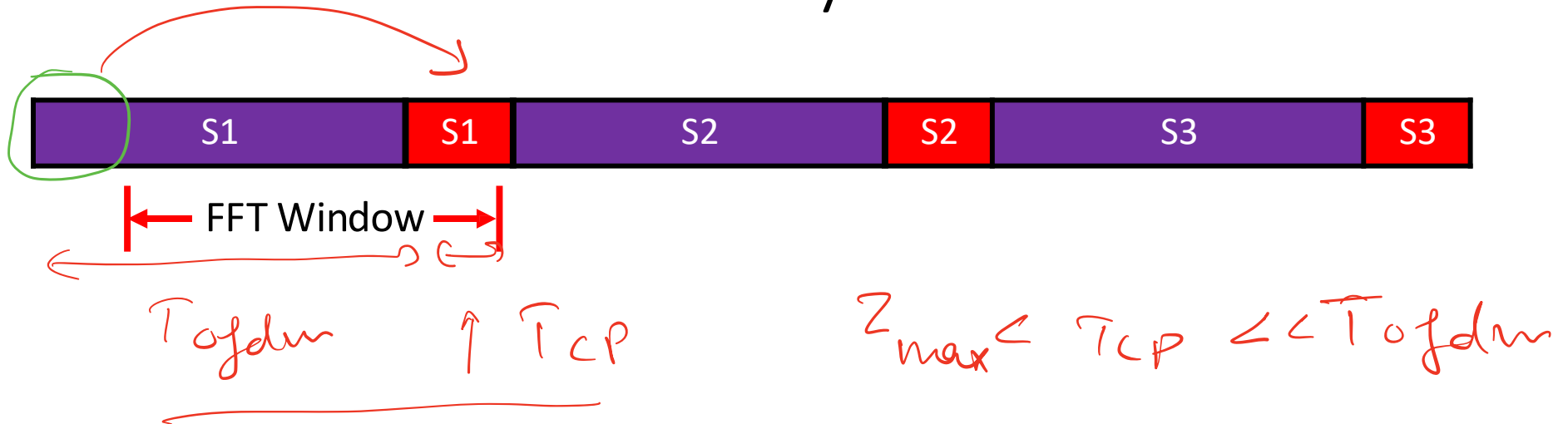


- DFT (FFT) assumes time samples are periodic of period N
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$$x[t - \tau \bmod N] \rightarrow X[f] e^{-j \frac{2\pi f \tau}{N}}$$

OFDM Cyclic Prefix



N samples
fft

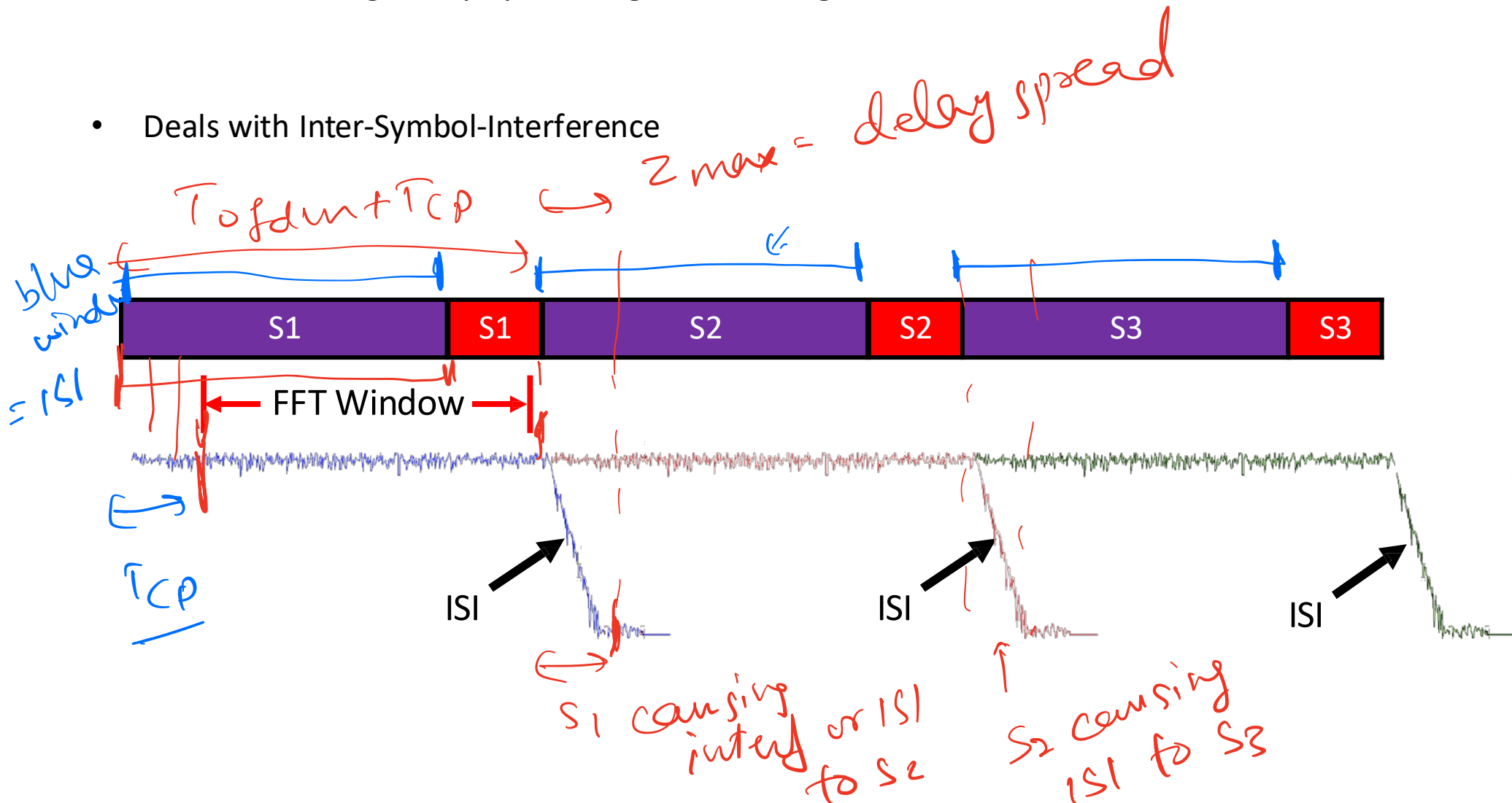
- Even if FFT window is misaligned, CP ensures that all samples come from the same symbol → Orthogonality is preserved!
- Simple Phase Shift → Can be corrected by lumping with channel $H[f]$
- Cyclic Prefix can be created by:
 - Take first few samples and append them to end of symbol.
 - Take last few samples and prefix them to beginning of symbol.

or

OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference

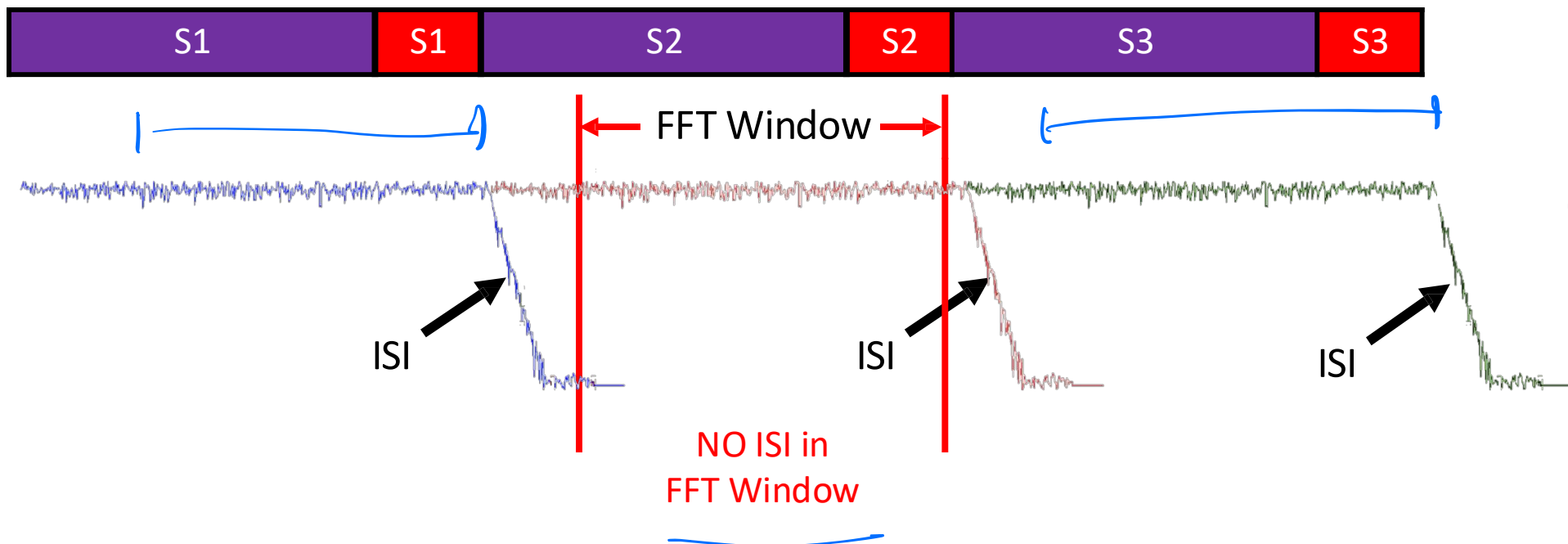


OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference

$$N_{\text{ofdm}} + N_{\text{cp}}$$

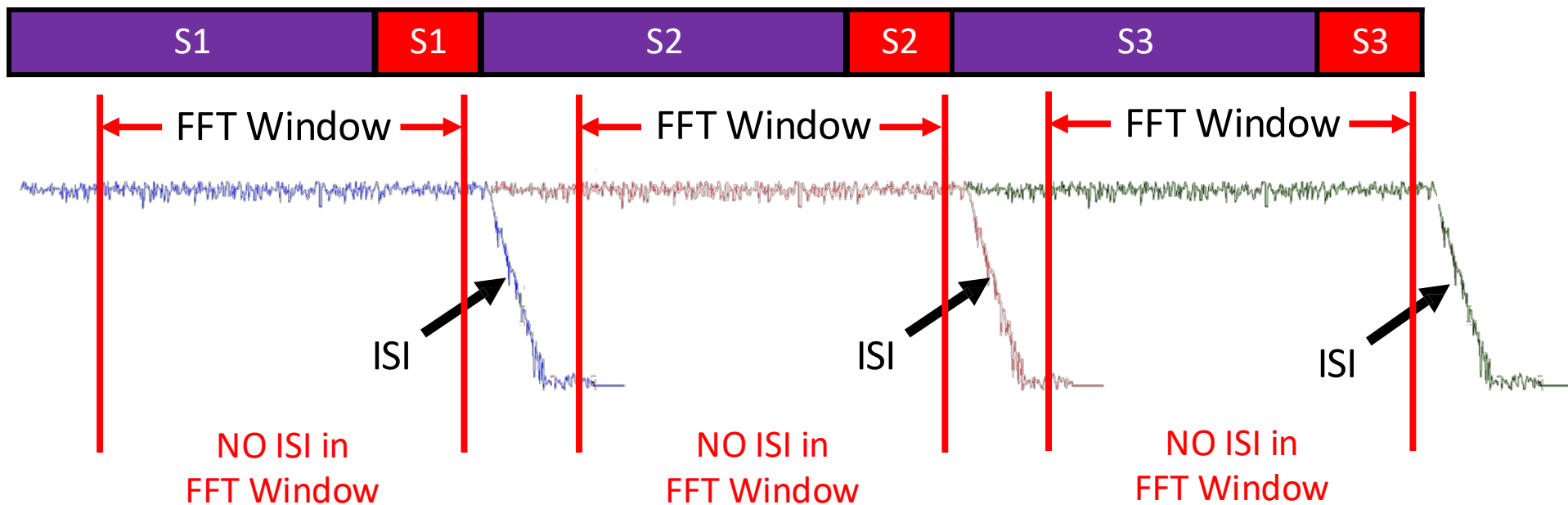


OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference

time →



OFDM Cyclic Prefix

Cyclic Prefix:

+ Preserves orthogonality by allowing some misalignment in FFT Window

+ Deals with Inter-Symbol-Interference

— Overhead: Send $CP + N$ samples for every N samples

$$\text{Overhead} = \frac{CP}{CP + N}$$

← extra overhead
← total duration of OFDM

e.g. WiFi 802.11n: $N = 64$, $CP = 16 \rightarrow \text{Overhead} = 20\%$

e.g. LTE: $N = 1024$, $CP = 72 \rightarrow \text{Overhead} = 6.5\%$

5G flexible

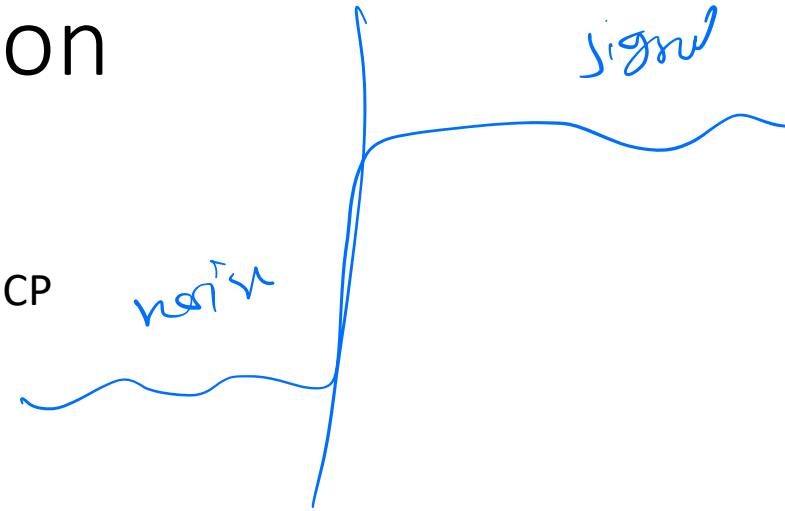
OFDM Cyclic Prefix



- Cyclic prefix is a not a bullet proof solution.
- Can still end up misaligned!
- Need a way to ensure we detect the beginning of the packet correctly.
- If we do, CP will ensure that even if we are not accurate, we can still decode.

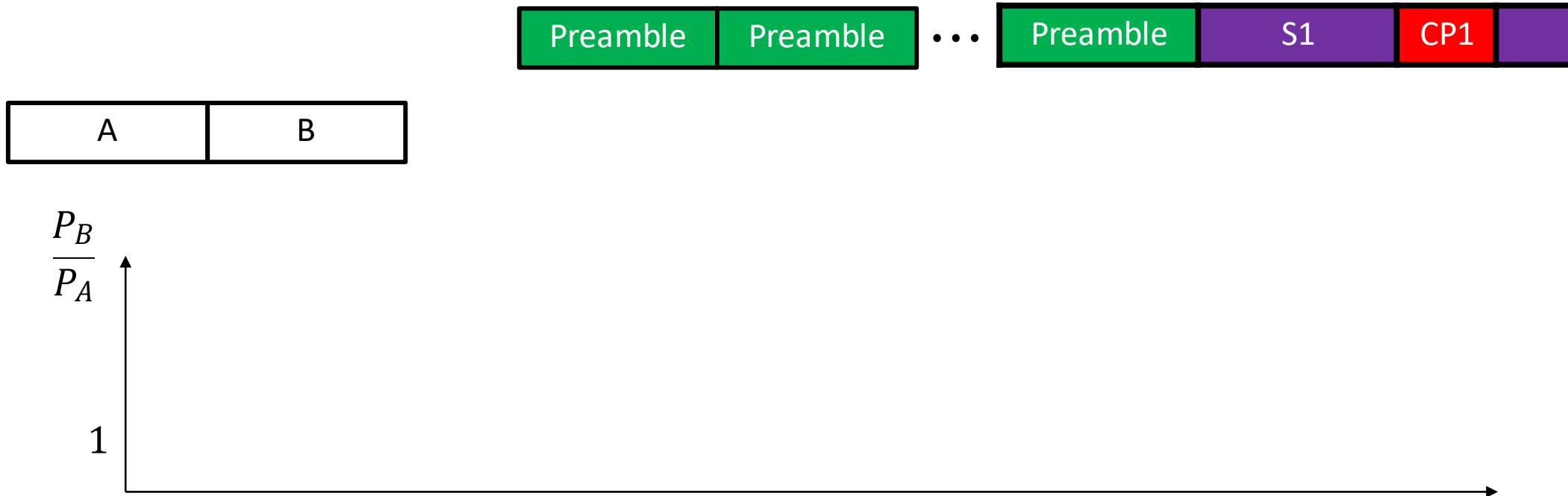
OFDM Packet Detection

- Detect Beginning of packet to make sure we are within the CP
- Send Training Sequence: Preamble Symbols
- Preamble Symbols: Known Symbol Repeated at the beginning of packet



- Generally, 2 copies of preamble (LTS – Long Training Sequence) used in WiFi + 0.5 copy as cyclic prefix

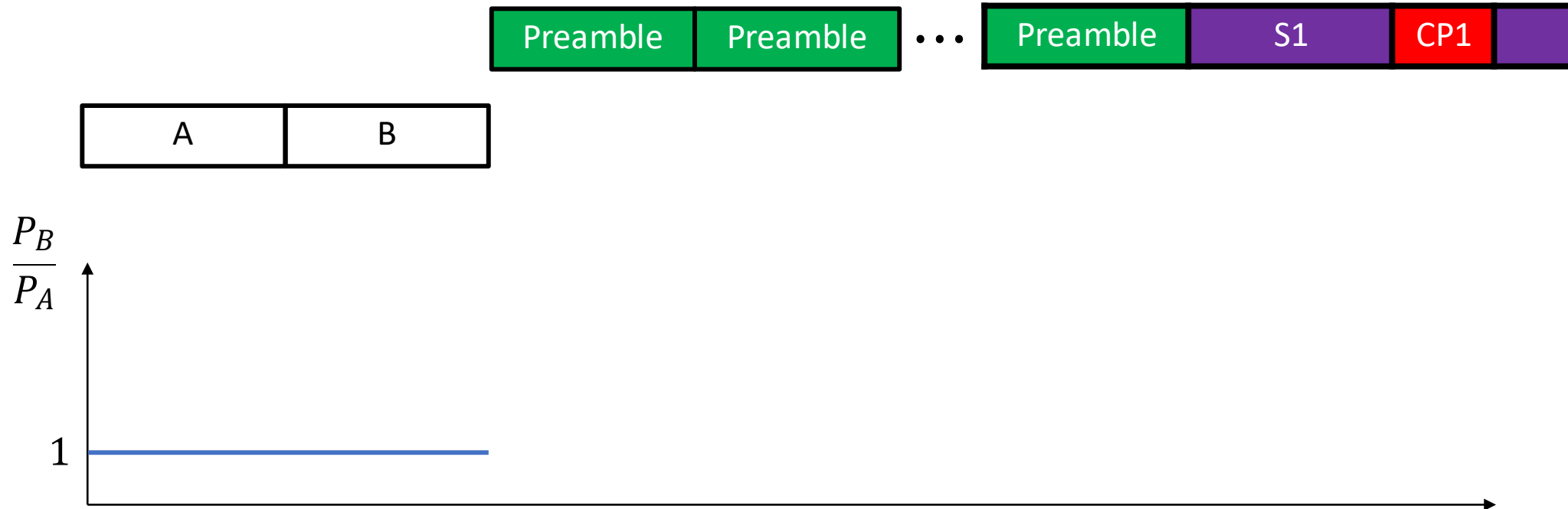
OFDM Packet Detection: Sliding Window



- Two windows of L ($2N$) samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

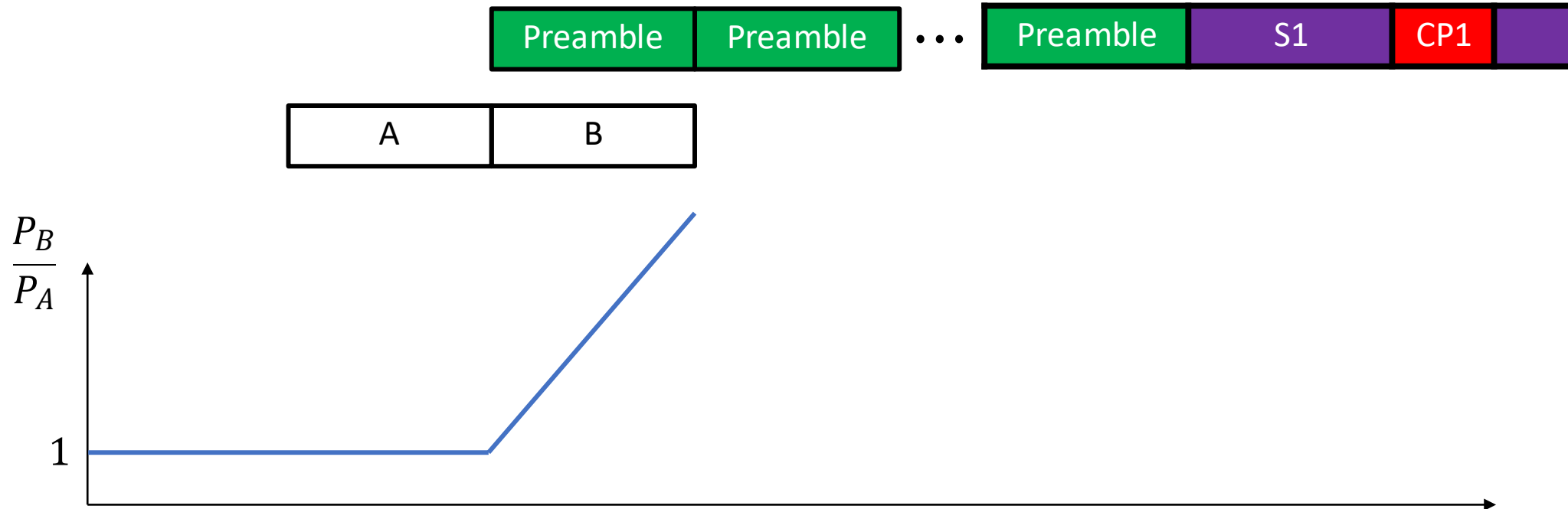
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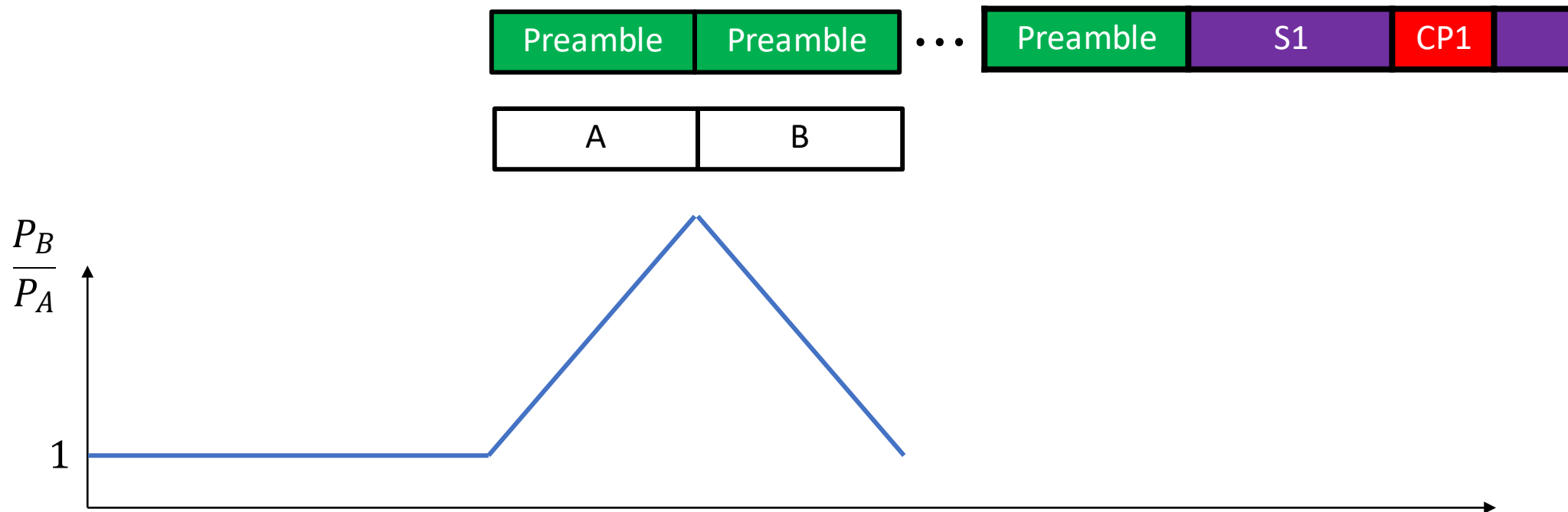
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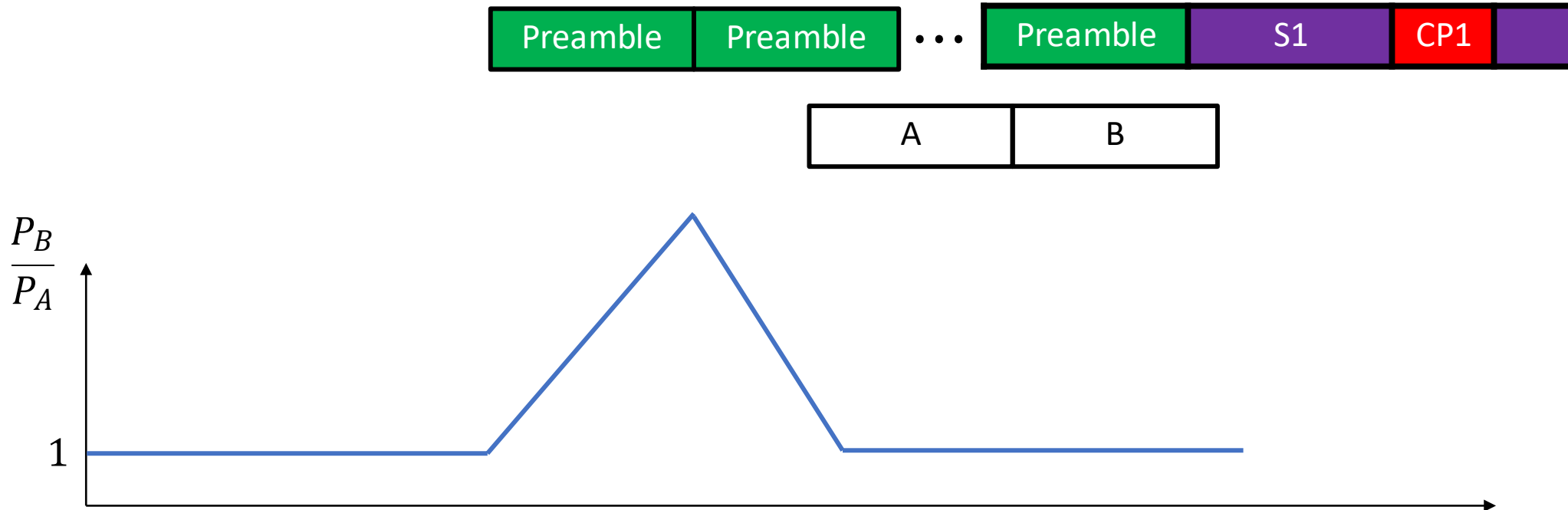
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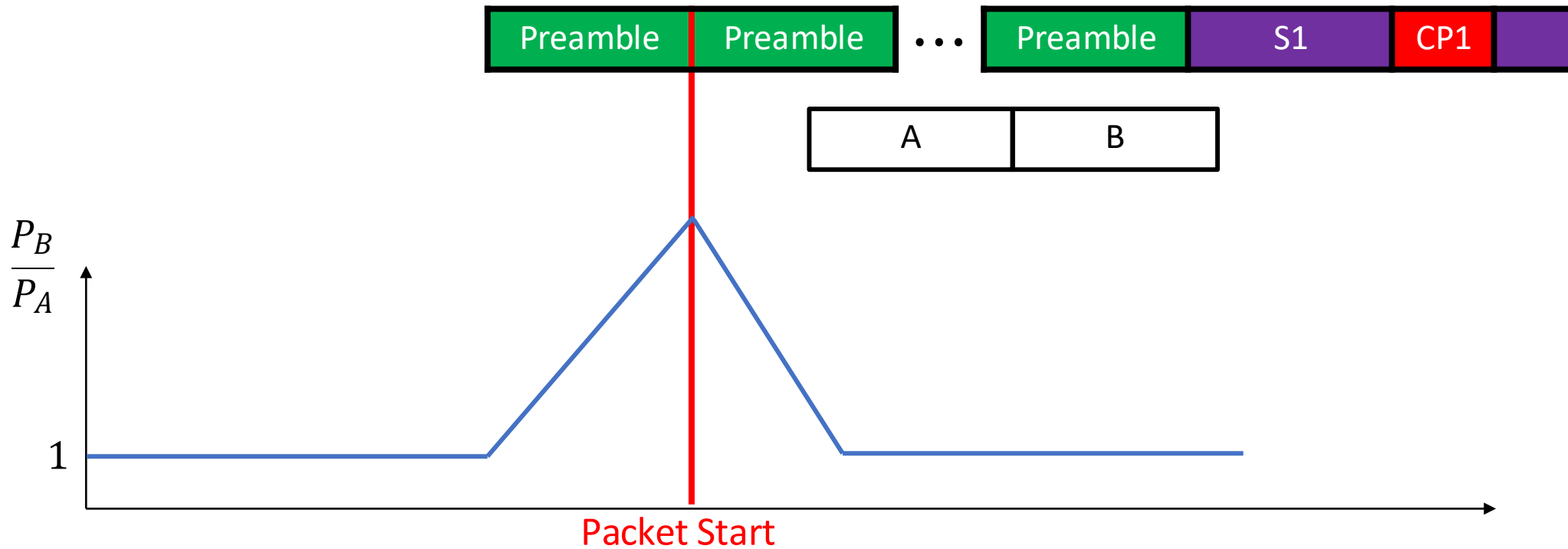
OFDM Packet Detection: Sliding Window



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OFDM Packet Detection: Sliding Window



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OFDM Packet Detection: Schmidl–Cox, 802.11 style (Reading Exercise - Piazza)

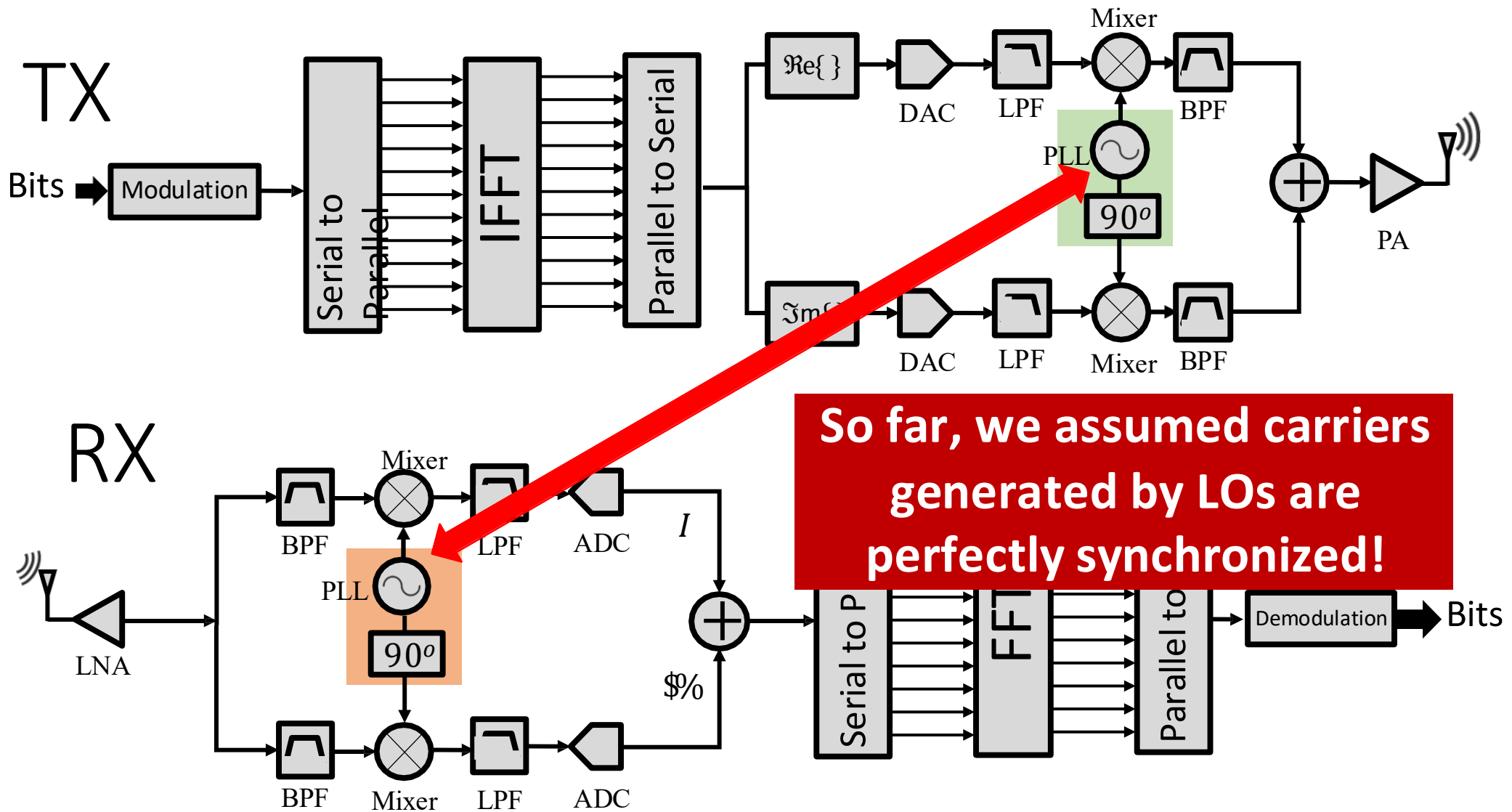
How often we send preamble over time?

Ans: Channel coherence time

WiFi Example: STS and LTS

OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers



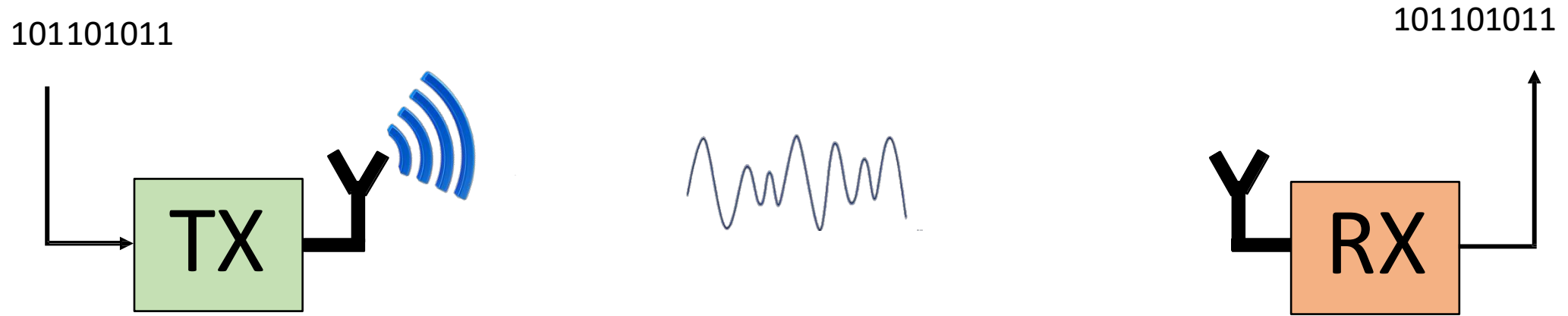
Carrier Frequency Offset



$$\begin{aligned}
 x(t) &\Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f_c t} \\
 &\Rightarrow h(t) * x(t) \\
 &\Rightarrow y(t) = h(t) * x(t) + v(t)
 \end{aligned}$$

Assumes TX & RX perfectly synched

Carrier Frequency Offset



$$\begin{aligned}
 x(t) &\Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f t} \times e^{j2\pi f' t} \\
 &\quad \text{TX \& RX are not synched} \\
 &\quad \text{CFO: } \Delta f_c = f_c - f'_c \\
 &\Rightarrow h(t) * x(t) e^{-j2\pi \Delta f_c t} \\
 &\Rightarrow y(t) = h(t) * x(t) e^{-j2\pi \Delta f_c t} + v(t)
 \end{aligned}$$

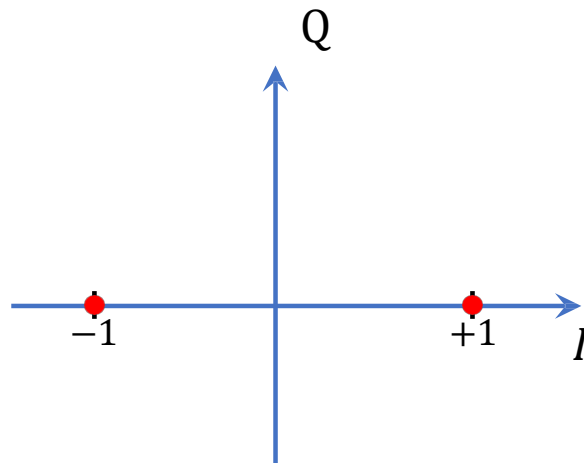
Phase changes with time!

Carrier Frequency Offset

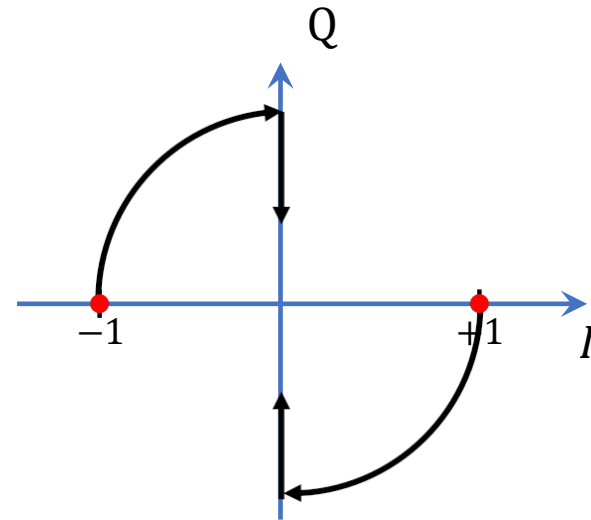
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$



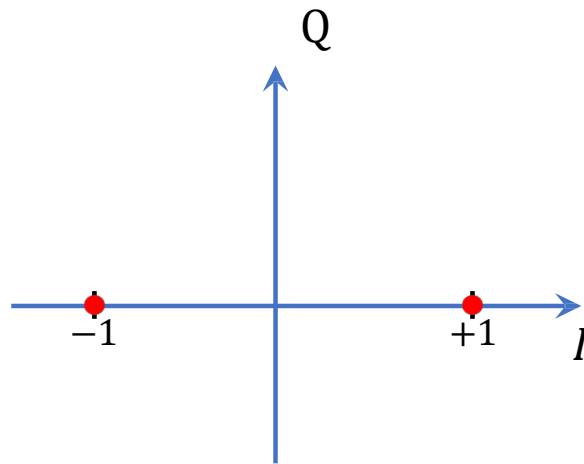
$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Carrier Frequency Offset

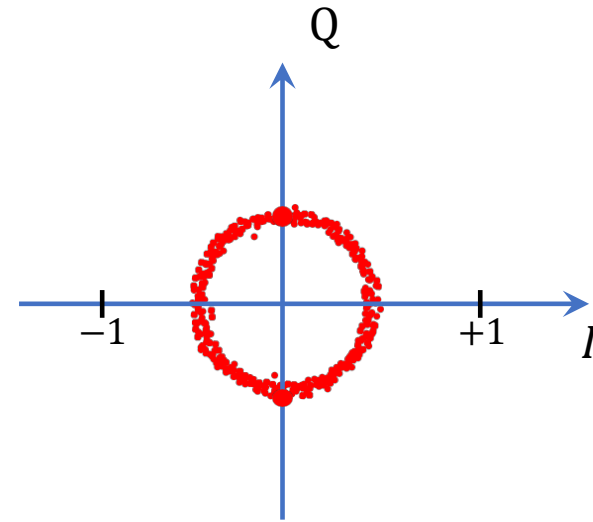
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

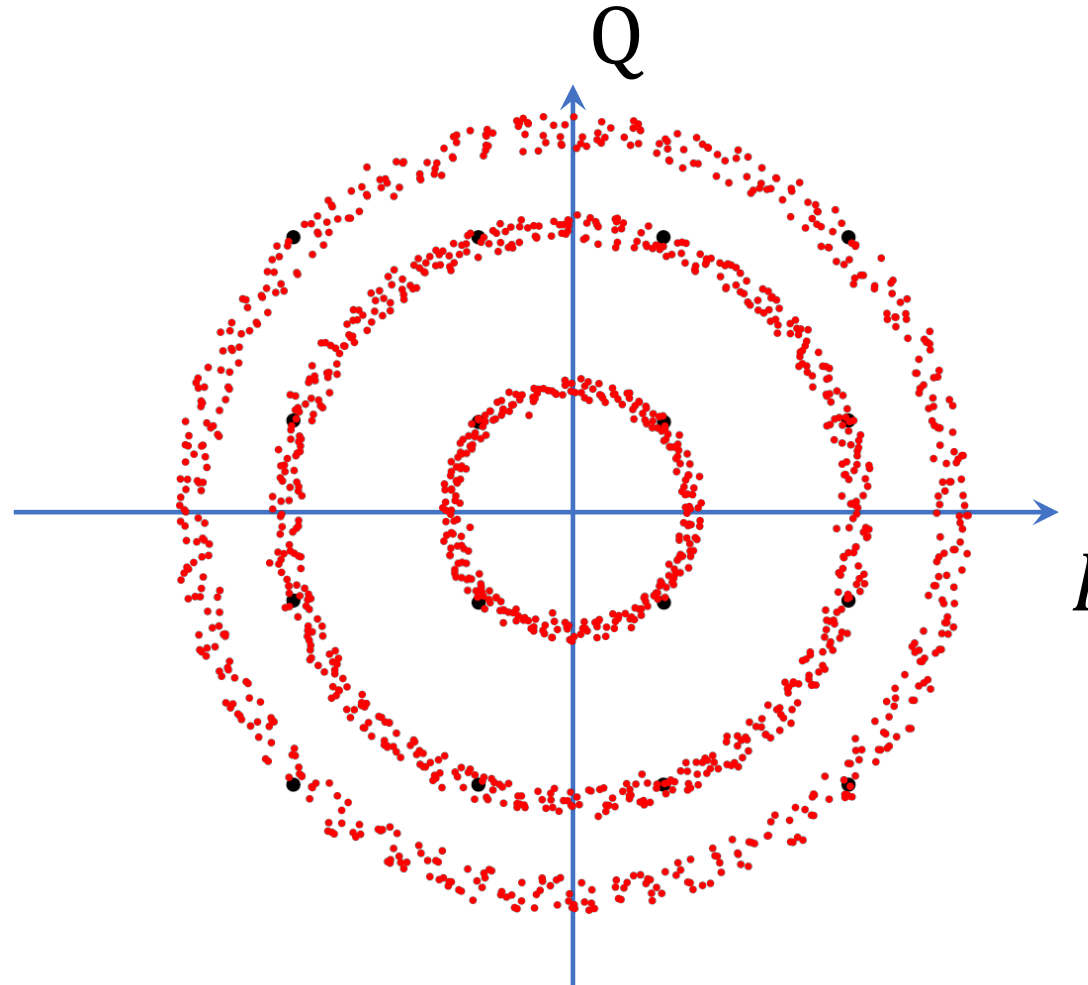


$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Impossible to Decode!

Carrier Frequency Offset

Consider 16 QAM Modulation



Need to estimate and correct CFO to decode!

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1(t) = x(t)e^{-j2\pi\Delta f_c t}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2(t) = x(t)e^{-j2\pi\Delta f_c (t + T)}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] = \sum_{t=1}^N x[n]x^*[n]e^{j2\pi\Delta f_c NT_s}$$

$$= e^{j2\pi\Delta f_c NT_s} \sum_{t=1}^N |x[n]|^2 \quad \rightarrow \quad \Delta f_c = \frac{\angle A}{2\pi NT_s}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] \quad \longrightarrow \quad \Delta f_c = \frac{\angle A}{2\pi nT_s}$$

- Correct CFO:
$$y[n] \times e^{j2\pi\Delta f_c nT_s}$$

OFDM Channel Estimation



- Use Preamble to estimate the channel

$$y(t) = h(t) * x(t) \leftrightarrow Y(f) = H(f) X(f)$$

- Send $X(f)$: $-1, +1, -1, -1, -1, +1, \dots$
- Receive: $-H(1), H(2), -H(3), -H(4), -H(5), H(6), \dots$

- Estimate:
$$\tilde{H}(f) = \frac{Y(f)}{X(f)}$$

- Use two preambles to average noise:
$$\tilde{H}(f) = \frac{Y_1(f) + Y_2(f)}{2 X(f)}$$

Phase Tracking

So Far: Estimated and Corrected For Coarse Value of CFO

- Residual CFO:

$$y(t) = h(t) * x(t) e^{j2\pi\Delta f_c t} + v(t)$$

$$\Delta f_c = df_c + \delta f_c$$

Coarse CFO

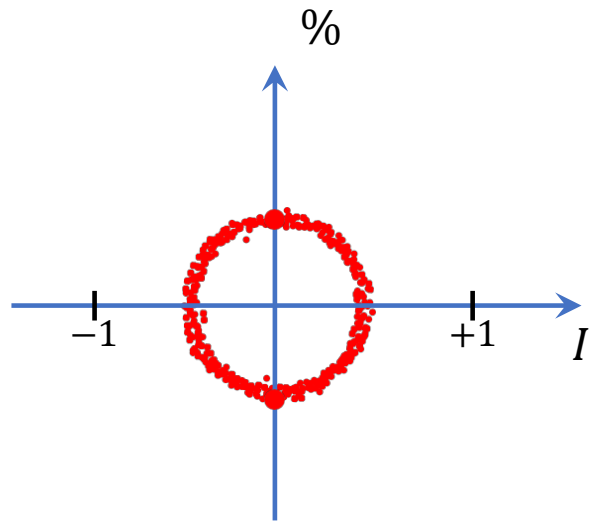
Residual CFO

We estimated and corrected for coarse CFO!

Even small residual can accumulate over time
to create large phase: $e^{-j2\pi\delta f_c t}$

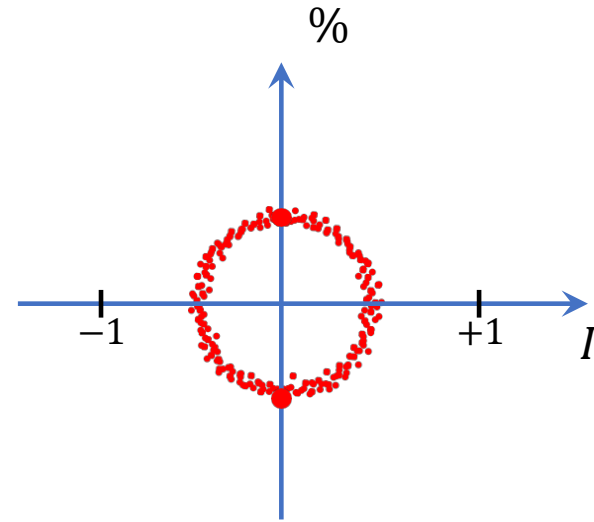
Need to track the phase

Phase Tracking



$$h x(t - r) e^{j2\pi\Delta f_c t} + v(t)$$

Without Course CFO Correction



$$h x(t - r) e^{j2\pi\delta f_c t} + v(t)$$

Without Residual CFO Correction

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

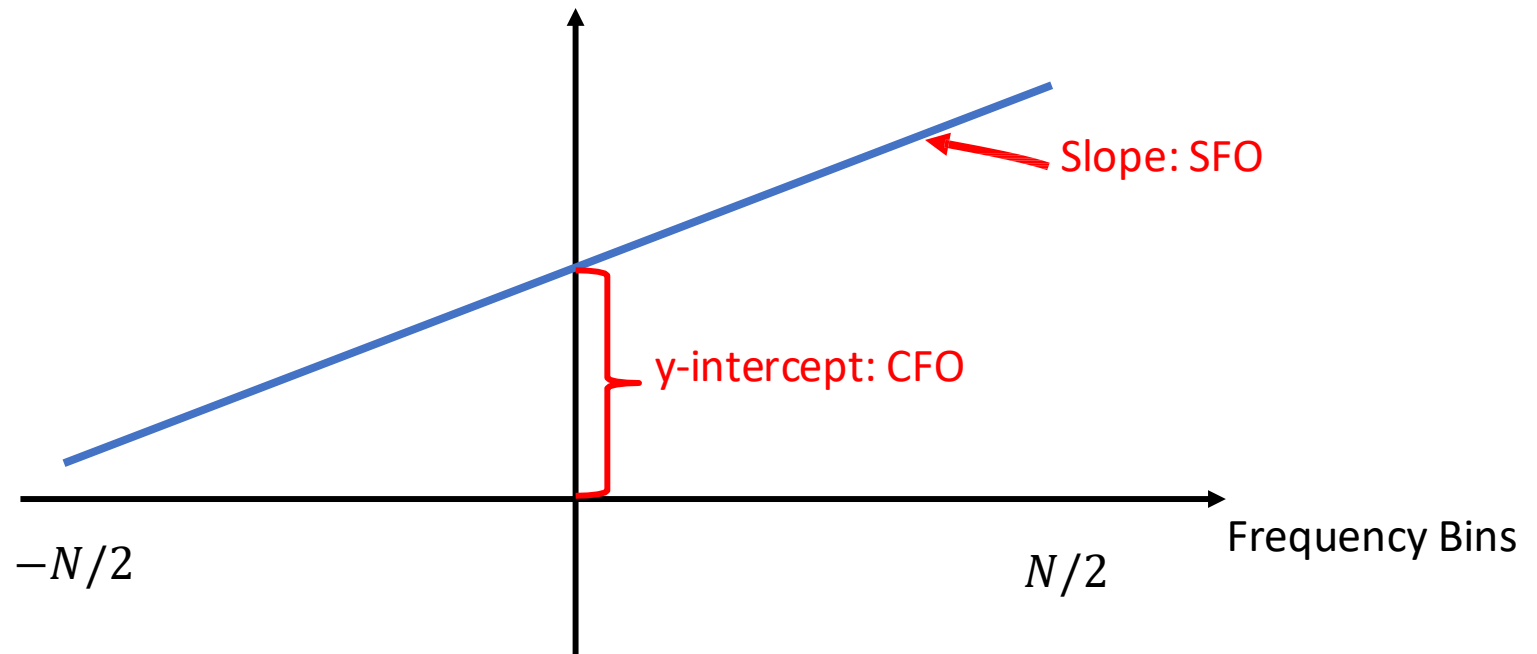
$$Y_1[f] = X_1[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

$$Y_2[f] = X_2[f] e^{j \frac{2\pi f (n+N+CP) \delta T_s}{N} - 2\pi \delta f_c (n+N+CP) T_s}$$

Phase accumulation: $\Delta\phi = 2\pi f \frac{(N+CP)\delta T_s}{N} - 2\pi \delta f_c (N+CP) T_s$

Phase Tracking

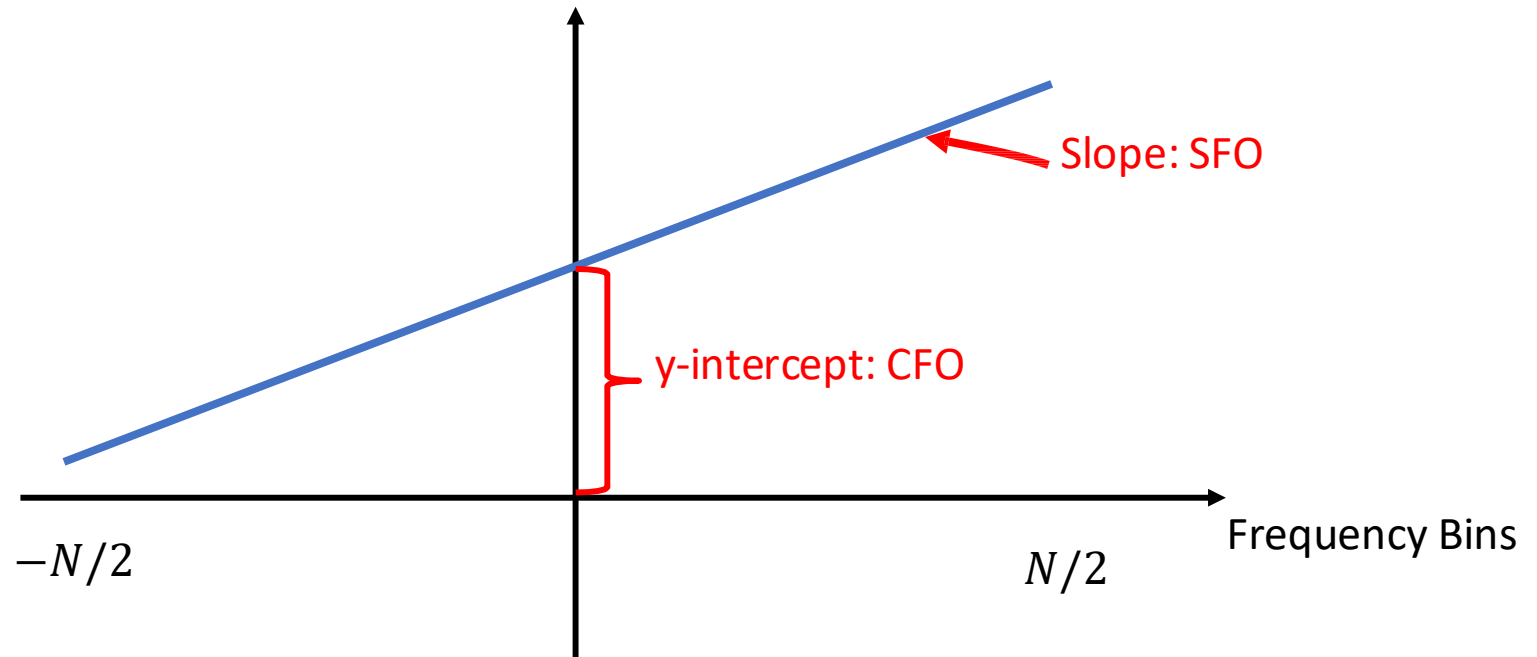
- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



OFDM Phase

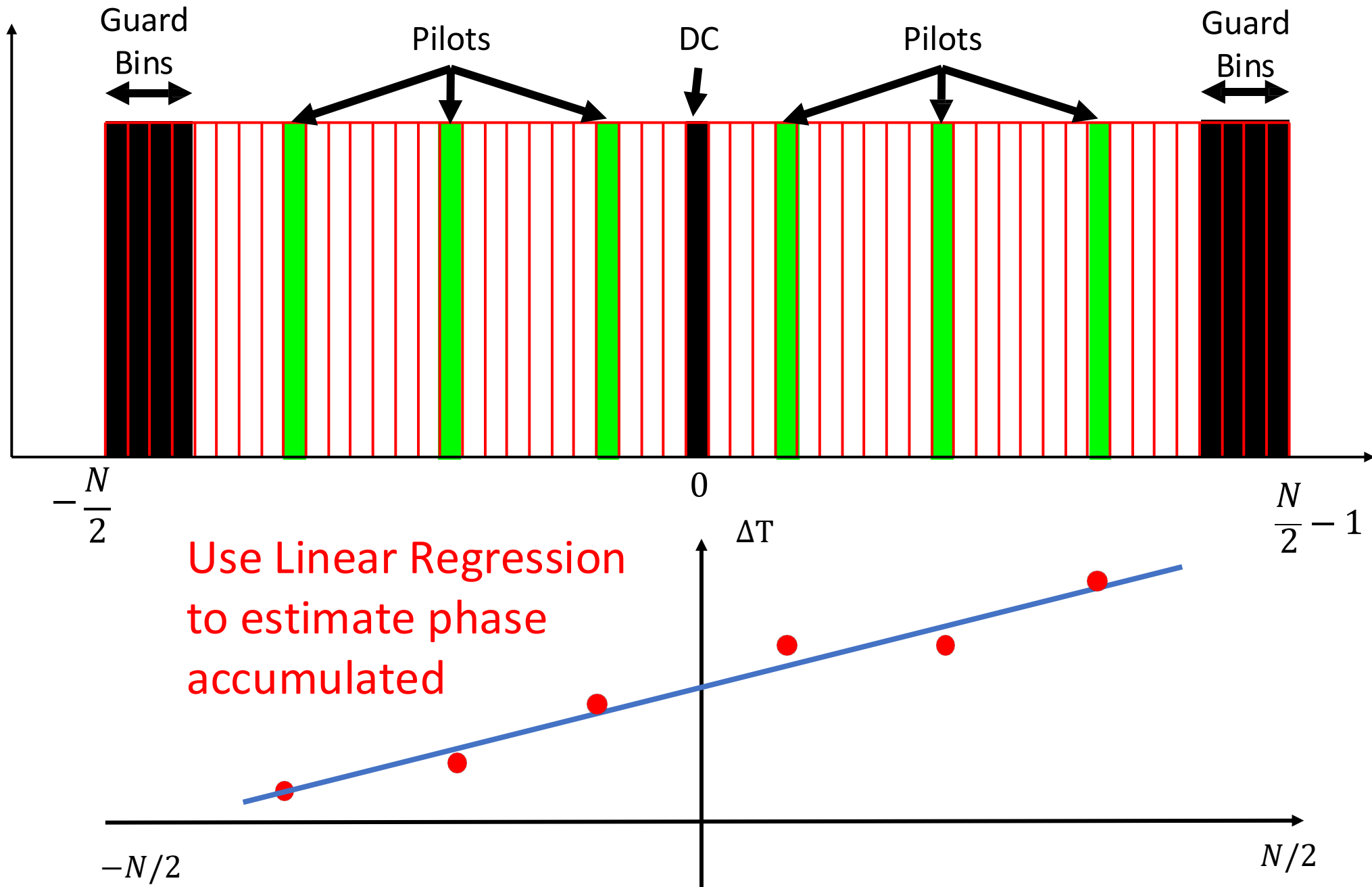
Tracking

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



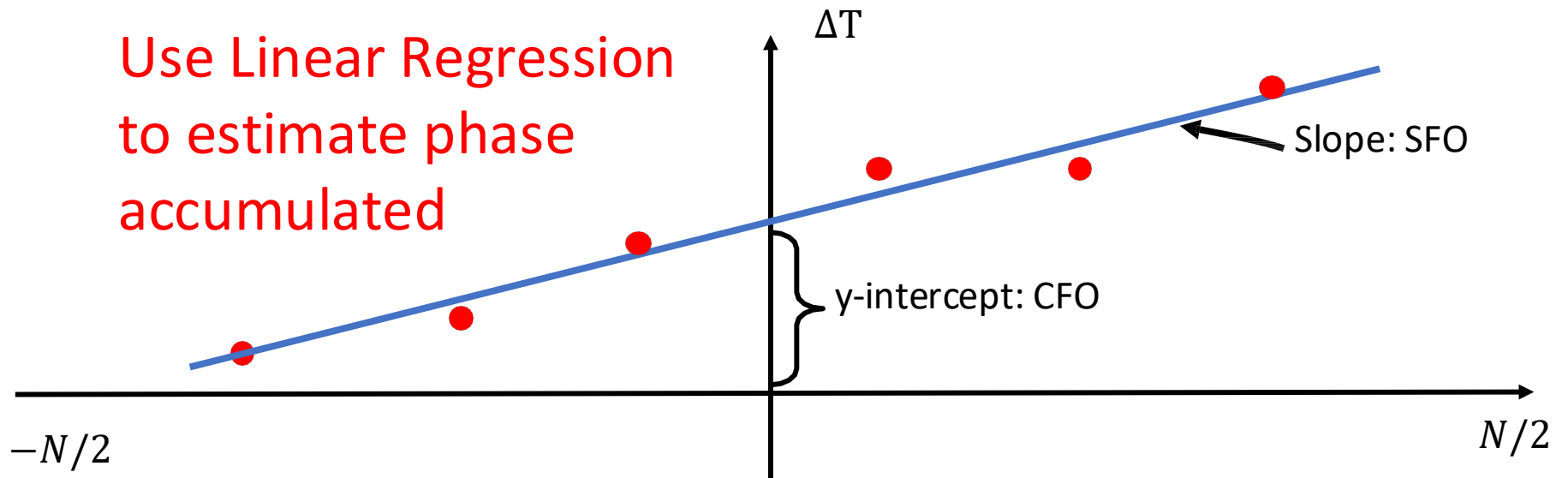
- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

OFDM Phase Tracking



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

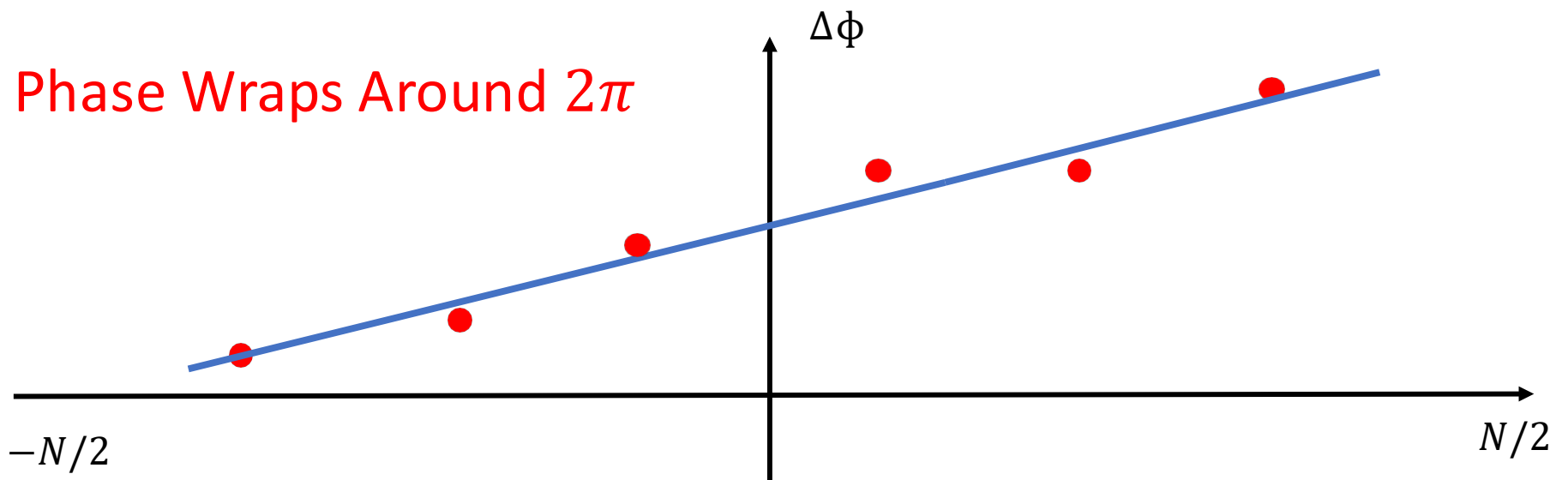


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

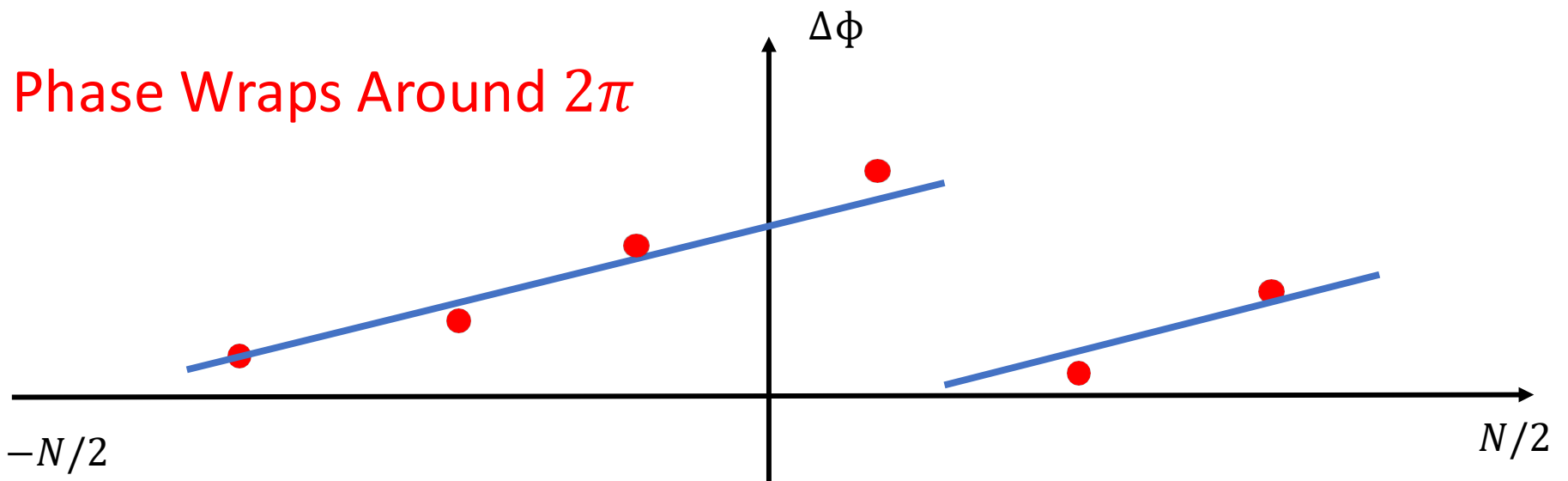
$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

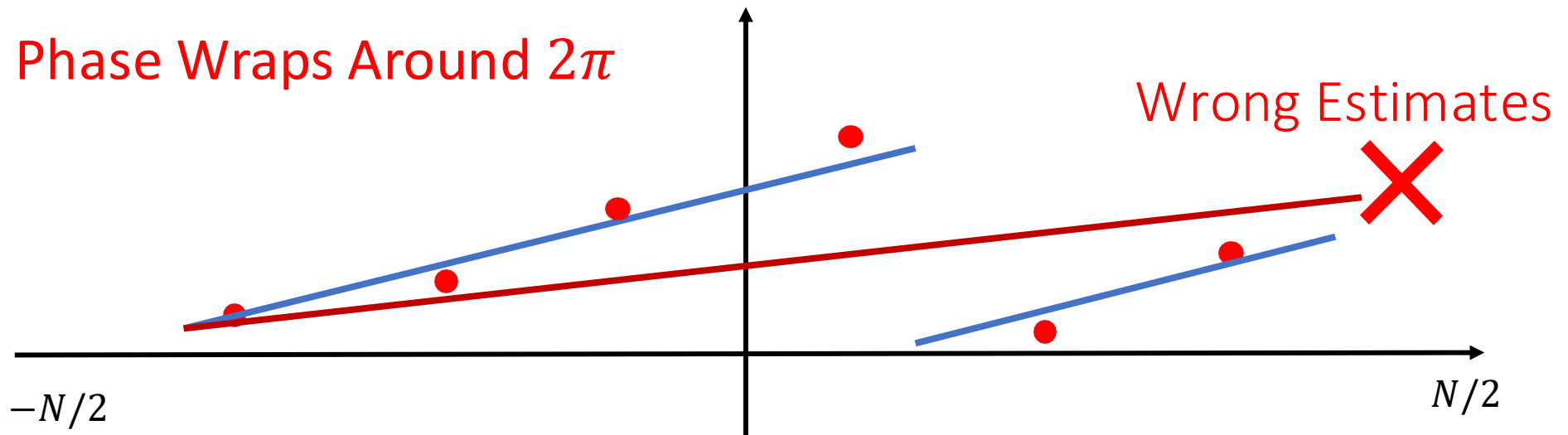
Be Careful!



Be Careful!



Be Careful!

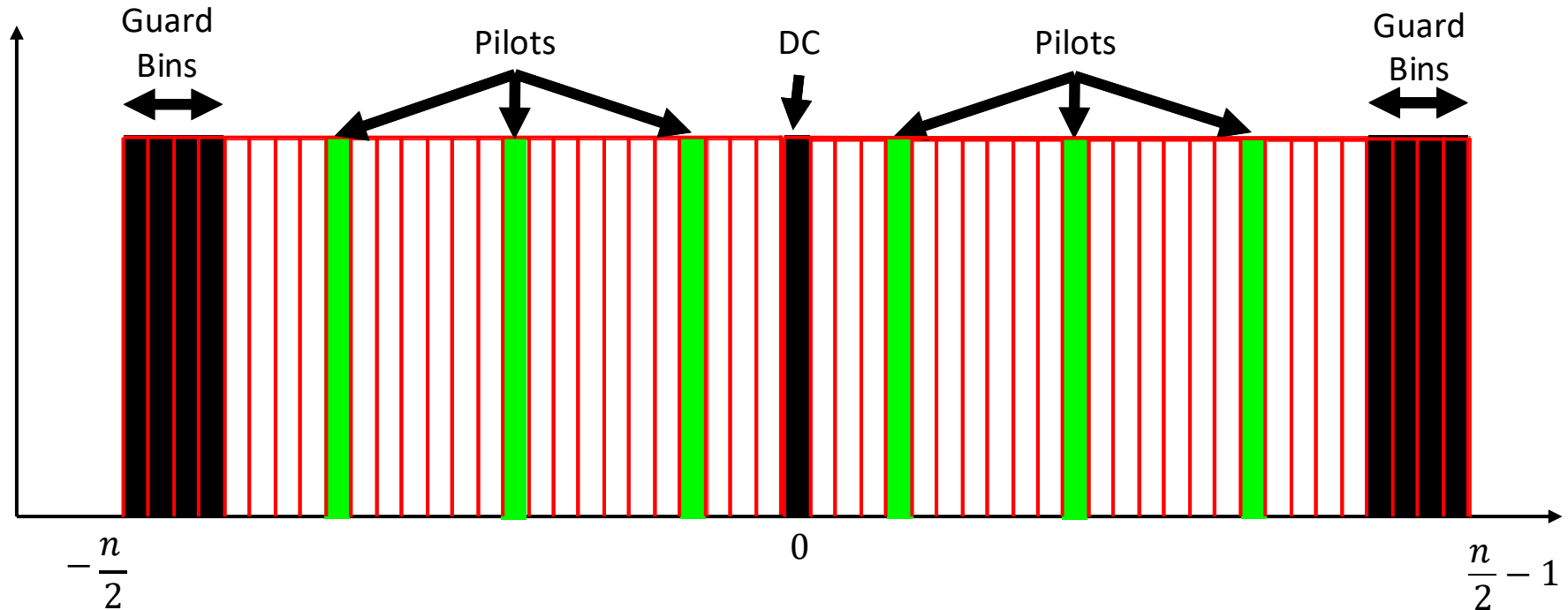


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

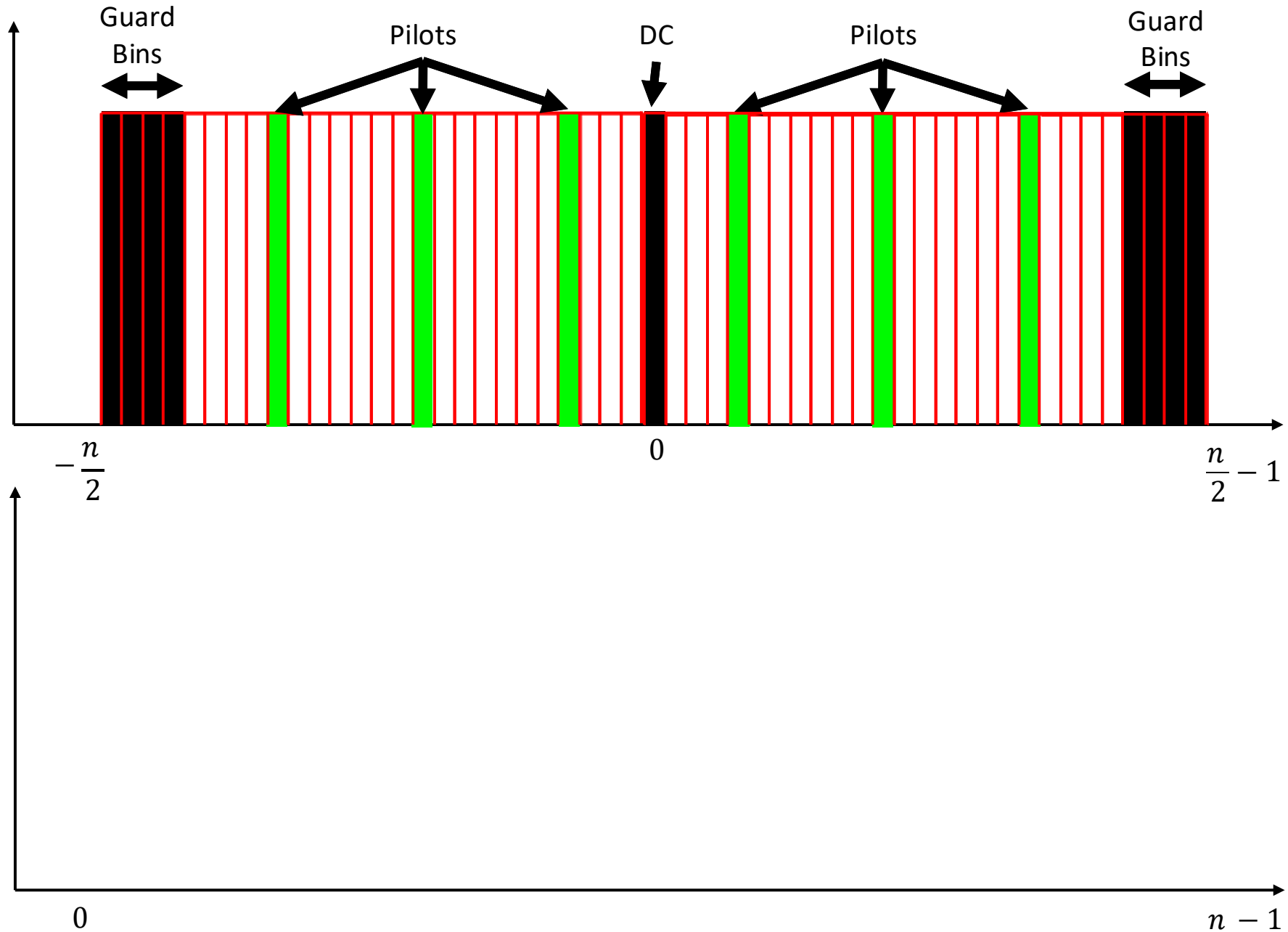
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



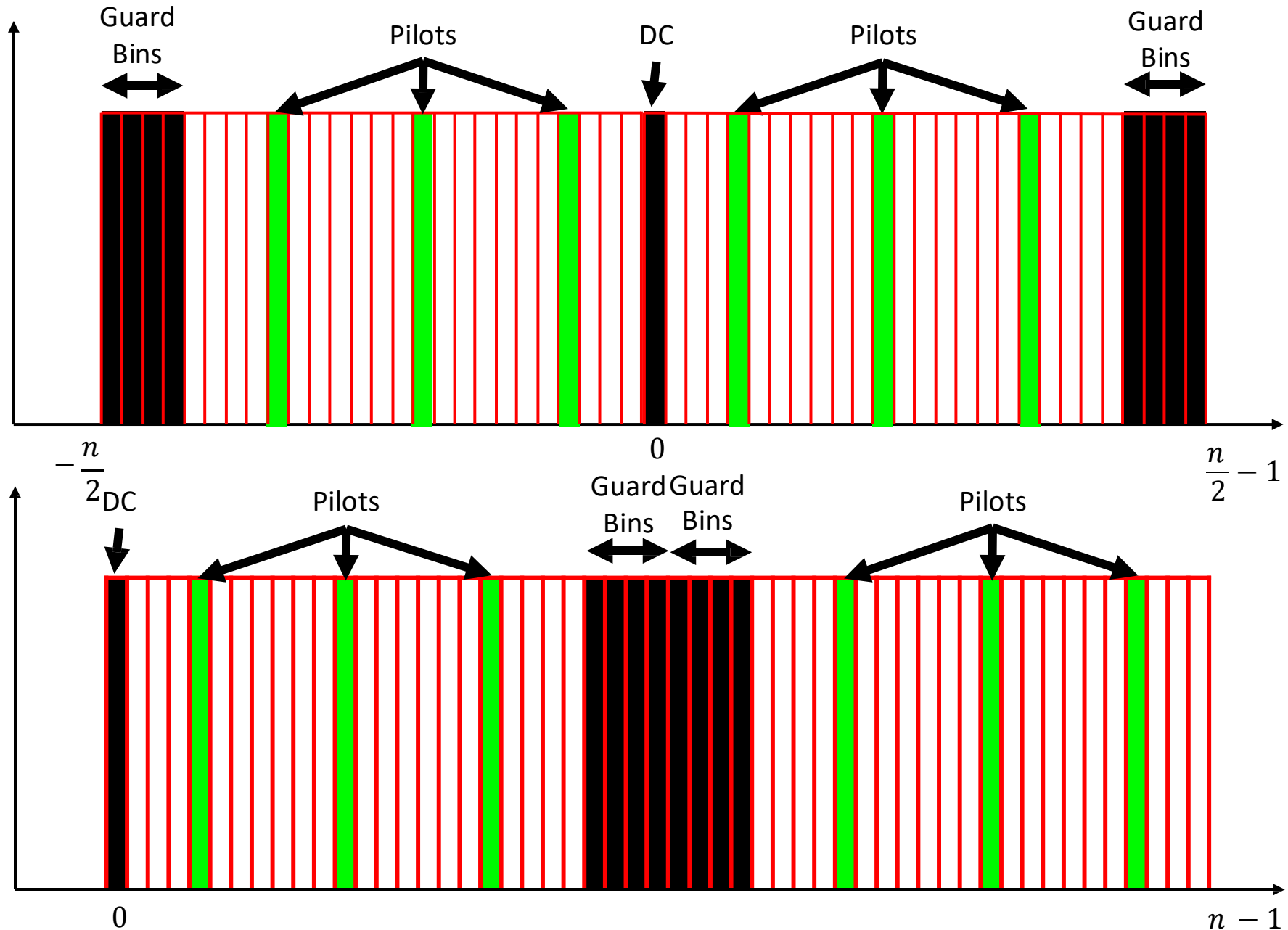
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



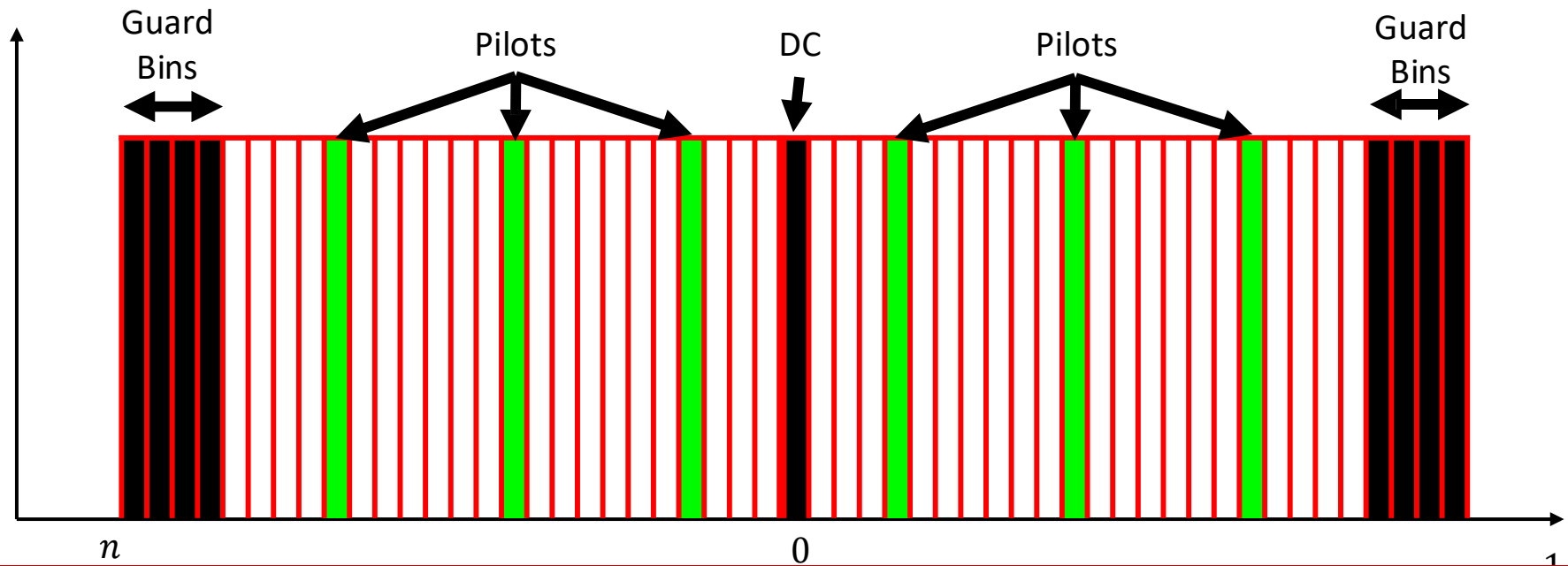
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$

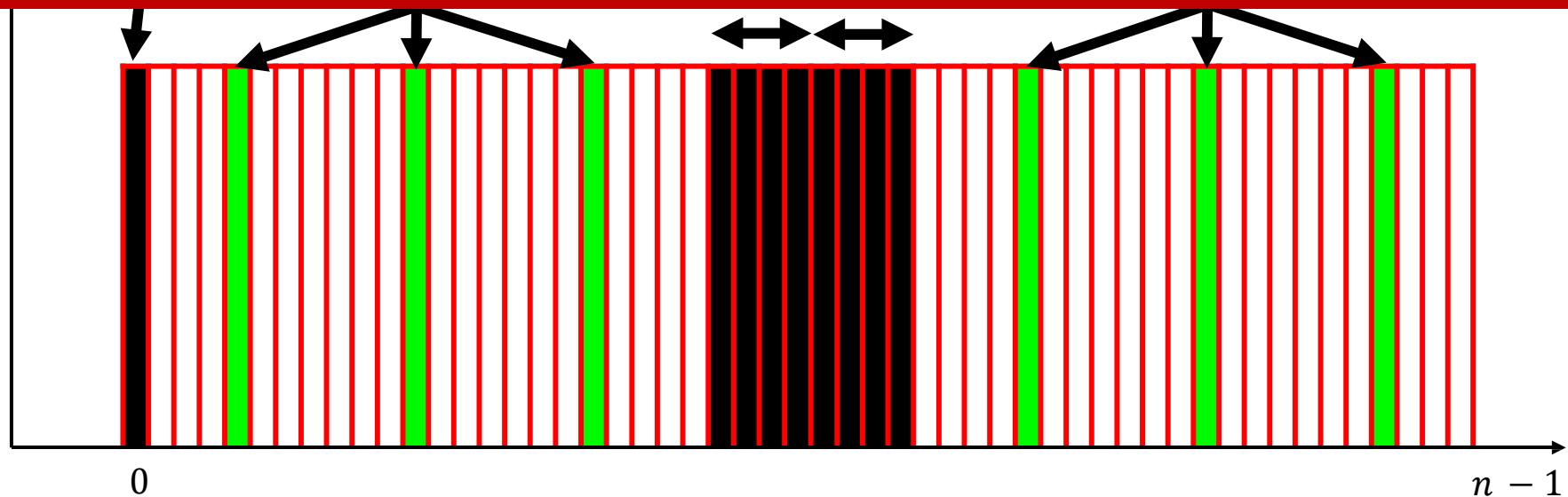


Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$



Need to `fftshift()` function in MATLAB + provided functions to convert indexing.



OFDM: Putting it Together

At TX:

- Create preamble symbol from training sequence (Uses BPSK)
- Repeat preamble symbol:
 - 4 times for packet detection
 - 2 times for CFO estimation
 - 2 times for channel estimation
 - Add CP for the last preamble
- Create data symbol from:
 - Data bits (Uses BPSK, QPSK, M-QAM)
 - Pilot bits (Uses BPSK)
- Add cyclic prefix to data symbols.

OFDM: Putting it Together

At RX:

- Detect beginning of packet.
- Estimate & correct for CFO.
- Jump $\approx 0.75 CP$ samples into symbol to avoid ISI
- Estimate the channel.
- For each subsequent data symbol:
 - Remove CP
 - Take FFT of Size N
 - Correct for channel by dividing with $\tilde{H}(f)$
 - Use linear regression to estimate residual CFO and SFO
 - Estimate accumulated phase $\Delta\phi(f)$ for each frequency bin
 - Add $\Delta\phi(f)$ to channel estimate $\tilde{H}(f)$
 - Decode Bits